Designing Consent: Choice Architecture and Consumer Welfare in Data Sharing

Chiara Farronato, Andrey Fradkin, Tesary Lin

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Abstract

We study how cookie-consent design affects consumer welfare using a framed field experiment that randomizes consent banners. We examine three margins: visual nudges, defaults when users dismiss banners, and whether consent is set at the website or browser level. Visual changes have little effect, but deliberate obstruction such as hiding "reject all" substantially shifts choices. Because many users close banners, welfare depends on both the website default and users' beliefs about defaults. We estimate a model of consumer choice. The model estimates show that when consent is requested on a site-by-site basis, the optimal design eliminates obstructions and aligns defaults with the majority preference. Relative to this optimum, typical U.S. consent practices cost 20% of surplus through obstruction, and typical EU consent practices decrease surplus by 6%. Correcting users' beliefs about website default raises surplus by 15% compared to the optimal site-by-site consent. The largest welfare gains, however, come from replacing site-by-site consent with browser-level consent, which helps consumers avoid the time costs of making repeated consent decisions.

Keywords: online privacy, choice architecture, consumer welfare, field experiments, consumer data

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All authors contributed equally to the paper. Chiara Farronato (cfarronato@hbs.edu): Harvard Business School, NBER, and CEPR. Andrey Fradkin (fradkin@bu.edu): Boston University and the MIT Initiative on the Digital Economy. Tesary Lin (tesary@bu.edu): Boston University. We thank Connor Haskin, Hayden Schrauff, and Chloe Song for excellent research assistance, and Audacious Software for software development. This work received generous support from the Internet Society Foundation and Boston University's Digital Business Institute.

1. Introduction

Consumer data is a key input into the digital economy, powering everything from algorithmic recommendations to targeted advertising. Yet the collection and use of this data have sparked privacy concerns, prompting regulators from the European Union to California to demand that firms obtain explicit user consent before data collection and use. In response, firms now regularly solicit consent, most commonly through cookie consent banners.

Data consent requirements are controversial for at least two reasons. First, firms routinely design interfaces that steer users toward broad data collection and use, potentially preventing consumers from making their preferred choice. These choice architectures, often labeled *dark patterns* in public discourse,[1] include deliberate obstruction (e.g., hiding rejection options behind extra clicks), reordering options to favor consent, or using differential visual salience. Second, requiring each firm to obtain user consent can result in frequent consent banners, which disrupts the user experience and may decrease decision quality due to "consent fatigue."[2] For these reasons, policymakers are considering simplifying consent requirements by letting consumers set their cookie preferences once at the browser level, rather than for each website.[3]

We study how three features of choice architecture shape cookie consent decisions and consumer welfare: 1) the design of consent interfaces, 2) default settings, and 3) the frequency of consent choices. To do so, we conduct a framed field experiment that randomizes consent banners as users browse the internet during the course of a week, allowing us to identify the causal effects of banner design. We find that different design features vary sharply in their effects on choices. Deliberate obstruction, such as removing "reject all cookies" from the main interface, substantially shifts choices, while purely visual manipulations (e.g., reordering or highlighting options) have little impact. In addition, many users simply close the banner, making both defaults and users' beliefs about those defaults consequential.

We then estimate a structural model of consumer choice to quantify welfare effects, allowing for heterogeneity in user preferences. Using data on interaction times with consent banners, combined with an assumption about the value of time, we express welfare in dollar terms under alternative consent policies. Under site-by-site consent, the optimal interface presents all options without obstruction and defaults to "accept all cookies" when users close the banner without making a choice. Removing obstruction

[1] https://www.deceptive.design/, accessed November 2025.

[2] https://www.marketplace.org/episode/2023/12/07/tired-of-trying-to-protect-your-data-privacy-youve-got-consent-fatigue

[3] For example, see https://www.politico.eu/article/europe-cookie-law-messed-up-the-internet-brussels-sets-out-to-fix-it/ and https://oag.ca.gov/privacy/ccpa/gpc.

generates the largest welfare gains, while changing defaults has more limited effects unless accompanied by correcting users' beliefs about defaults. Moving beyond site-by-site consent, a browser-level consent mechanism delivers substantially larger welfare gains than even the optimal banner design, as it eliminates the time costs of repeated interactions. These gains hold across users with differing preferences, whom we classify as acceptors, rejectors, and discerners. Overall, browser-level controls can increase welfare by up to 124% compared to the optimal site-by-site design, and by at least 36% even under more conservative assumptions.

Our experiment uses Cookie Manager, a customized browser extension built on the Webmunk framework (Farronato, Fradkin and Karr 2024), to randomize consent interfaces across users and web domains as participants browse the web. The interfaces vary the ease of selecting among three options: accept all cookies, reject all (non-essential) cookies, and customize preferences by purpose. We design these interfaces to test three types of choice architecture. *Deliberate obstruction* removes an option from the main banner, requiring additional clicks to select it. *Reordering* emphasizes an option by placing it at the top of the list. *Differential highlighting* increases the salience of an option through color differences. The extension enforces participants' decisions whenever technically feasible, making user choices incentive-compatible. We use Prolific to recruit U.S. consumers who consent to install Cookie Manager.

The study includes two phases. In the first phase (*survey browsing*), we direct participants to visit specific websites. This structured browsing allows us to evaluate users' privacy preferences across the same set of websites. In the second phase (*organic browsing*), we observe participants' natural browsing behavior for a week. In both phases, we randomize consent interface designs as users encounter new web domains.

We find that under the *baseline interface*, with no obstruction or visual manipulations, most users accept all cookies. Deliberate obstruction has a large impact on choices, whereas visual manipulations have minimal effects. Hiding the "reject all cookies" option reduces rejection rates by 13 percentage points (a 68% decline). This strong effect is consistent with real-world practice, where obstruction is the most common design strategy (Utz et al. 2019). By contrast, reordering options or graying out alternatives increases consent rates by only 3 percentage points.

Closing the banner without making a selection is the second most common choice during organic browsing, after accepting all cookies. However, most participants incorrectly believe that doing so results in rejecting cookies, even though websites in the U.S. are typically not required to reject by default and often default to acceptance in practice. This mismatch creates a gap between intended and actual data-sharing outcomes, which we explicitly incorporate in our welfare analysis.

Consent interfaces also impose significant time costs. On average, participants spend 7.34 seconds per baseline interface during organic browsing (2.46 seconds at the median). When users click into the "settings" menu, which is required when preferred options are hidden, the interaction time increases by over 50%.

To quantify welfare implications, we estimate a model of consent decisions that accounts for consumer heterogeneity through a distribution of latent user types. The model captures this heterogeneity with three groups: users who mostly accept cookies, users who mostly reject them, and users who tailor their choices across websites. Two challenges arise for welfare measurement: mapping incorrect beliefs into choice mistakes and translating utility into dollar terms. To tackle the first challenge, we use survey data on participants' beliefs about the consequences of closing the banner and incorporate these beliefs directly into the surplus calculation following Train (2015). To tackle the second, we assume that the disutility from interacting with consent banners is proportional to time spent, and convert this into monetary terms using each user's implied hourly wage based on their reported income.

Our welfare analysis compares a range of existing and proposed consent policies. Within site-by-site regimes, the optimal design removes obstruction and defaults to "accept all," matching the majority preference. Relative to this optimum, current U.S. practices cost $0.6 per user-week in surplus (20%). In comparison, EU practices decrease consumer surplus by less (6%) by deviating from the optimal default. Correcting users' beliefs about the default when closing a banner increases consumer surplus by $0.43 per user-week (15%).

An alternative approach replaces site-by-site consent with browser-level privacy settings, commonly known as Global Privacy Control (GPC). This approach delivers substantially higher welfare, increasing surplus by $3.67 per user-week relative to the optimal banner design. This corresponds to a 124% gain, and at least 36% under more conservative assumptions. These gains are driven by eliminating the time costs of repeated consent decisions and hold even for users with heterogeneous preferences across websites. Consistent with this, 62.4% of participants report preferring browser-level controls to interacting with banners on each site.

These findings speak directly to the current policy debate on choice architecture related to online privacy. In the EU, both the Digital Services Act (DSA) and the Artificial Intelligence (AI) Act regulate choice architecture;[4] in the U.S., several states and the Federal Trade Commission have taken action against deceptive interface designs.[5] Our

[4]https://www.europarl.europa.eu/RegData/etudes/ATAG/2025/767191/EPRS_ATA(2025)767191_EN.pdf

[5]The Federal Trade Commission has fined Epic Games and Amazon for user interface designs that induce accidental purchases and obstruct subscription cancellation (https://www.ftc.gov/news-events/news/press-

results support these efforts: designs that deliberately obstruct choices reduce welfare and undermine users' ability to make choices consistent with their preferences. However, our analysis also suggests that reducing the frequency of decisions may yield even larger welfare gains. Browser-level consent mechanisms, already implemented by Mozilla and supported by the California Privacy Rights Act, could achieve these benefits while preserving user choice.[6]

Our work contributes to the growing literature on the economics of privacy, particularly studies that measure privacy preferences (Lin 2022; Collis et al. 2021; Ke and Sudhir 2023; Tomaino, Wertenbroch and Walters 2023; Tang 2023; Que 2017; Acquisti, John and Loewenstein 2013) and examine the impact of data collection practices on consumers (Goldfarb and Tucker 2011; Miller and Tucker 2018; Zhao, Yildirim and Chintagunta 2021; Bian et al. 2023) and firms (Johnson, Shriver and Goldberg 2023; Johnson et al. 2024; Goldberg, Johnson and Shriver 2024; Deisenroth et al. 2024; Wernerfelt et al. 2025). Importantly, we highlight that privacy choices in our setting are shaped not only by consumers' preferences over data sharing but also by their perceived benefits from cookie-based personalization and functionality. Our contribution lies in measuring these choices in a naturalistic, real-world setting and using them to estimate consumer surplus under alternative data-collection regimes, thereby offering a policy-relevant view of privacy trade-offs.

Our analyses relate to the recent theoretical work examining the economics of information markets and privacy (Bergemann and Bonatti 2019, 2024; Chen 2025). A central theme in this literature is the presence of externalities, as individual data-sharing decisions influence the privacy and welfare of others (Acemoglu et al. 2022; Rhodes and Zhou 2024; Miklós-Thal et al. 2024) by affecting firms' inference abilities. In this work, we abstract away from these spillover effects and instead focus on individual choices, in line with existing regulatory frameworks (Goldfarb and Tucker 2024).

We also contribute to the broader literature on choice architecture, particularly in the context of privacy decisions. Most empirical work has focused on documenting the prevalence of manipulative consent designs online (Mathur et al. 2019; Di Geronimo et al. 2020; Nouwens et al. 2020; Warberg et al. 2023). Attempts to quantify their effects on user behavior have typically relied on lab experiments or artificial settings (Acquisti, John

releases/2023/03/ftc-finalizes-order-requiring-fortnite-maker-epic-games-pay-245-million-tricking-users-making and https://www.ftc.gov/news-events/news/press-releases/2023/06/ftc-takes-action-against-amazon-enrolling-consumers-amazon-prime-without-consent-sabotaging-their). U.S. states such as California, Colorado, and Connecticut have enacted privacy regulations that ban companies from manipulating interface designs to increase data collection. (https://insightplus.bakermckenzie.com/bm/technology-media-telecommunications_1/united-states-consumer-protection-regulators-set-sights-on-dark-patterns.).

[6]See: https://blog.mozilla.org/netpolicy/2021/10/28/implementing-global-privacy-control/; https://globalprivacycontrol.org/press-release/20201007.html

and Loewenstein 2013; Utz et al. 2019; Luguri and Strahilevitz 2021; Habib et al. 2022; Lin and Strulov-Shlain 2023; Bielova et al. 2024; Baviskar et al. 2024). A few exceptions, such as D'Assergio et al. (2022), Posner et al. (2023), and Müller-Tribbensee, Miller and Skiera (2024), examine persuasive design in real-world contexts. Our study extends this literature by capturing privacy choices made during everyday browsing, thus offering greater real-world relevance. In addition, our analysis covers a wide range of websites, allowing us to assess how design manipulations interact with domain familiarity and popularity.

Finally, our work is related to the existing literature on behavioral biases and their implications for firm competition (Huck and Zhou 2011; Spiegler 2014; Ho, Hogan and Scott Morton 2017; Decarolis, Li and Paternollo 2023). This literature examines how factors such as switching costs and obfuscation strategies can limit competition in product markets. Our findings extend this analysis to the domain of data collection strategies. Interestingly, we find that the effectiveness of choice architecture is uniform across websites, regardless of popularity. This suggests that policies targeting choice architecture might not reduce incumbents' data advantage (Campbell, Goldfarb and Tucker 2015; Aridor et al. 2024; Johnson et al. 2024). Although not the primary focus of our study, if the marginal value of data is higher for smaller firms, such policies may even weaken a potential leveling mechanism (Hagiu and Wright 2023), to their disadvantage.

The rest of the paper is structured as follows. Section 2 presents the institutional context and our experimental design. We discuss our reduced-form findings in Section 3, and our model of user privacy preferences in Section 4. Section 5 compares the welfare impacts of policies regulating online consent. Section 6 discusses the implications and limitations of our findings.

2. Study Design

In this section, we introduce the institutional context to motivate our study, then present the consent interfaces in our experiment, the study phases and randomization, and describe the sample of participants.

2.1. Institutional Context

Our consent interfaces build on prior work that documents how firms use choice architecture to steer data-sharing decisions (Habib et al. 2022; Bielova, Santos and Gray 2024). We distinguish between two features. First, *design manipulations* affect how choices are presented. These include obstruction tactics that create friction around privacy-protecting

options (e.g., hiding "reject all" behind additional clicks), and visual manipulations that alter salience through layout, color, or ordering.[7] Second, *default settings* determine the outcome when users do not make an explicit choice, most notably when they close the banner.

Consent choice architectures are widespread. For example, Utz et al. (2019) find that 57.4% of major EU websites employ at least one manipulative element, and Nouwens et al. (2020) find similar patterns for over 80% of the top 10,000 UK websites. More recent evidence suggests that 85% of consent banners remain non-compliant with GDPR, with obstruction being the most common pattern (Nouwens et al. 2025). Similar practices arise in the U.S., where fewer than half of e-commerce websites display both "accept all" and "reject all" options on the main screen (Li 2022).

Regulations on consent choice architecture vary across jurisdictions. In the EU, the GDPR and the DSA explicitly restrict manipulative designs, requiring consent to be given through a "clear affirmative action" and prohibiting practices that make rejection harder than acceptance. EU regulations also favor privacy-protecting defaults, with many implementations defaulting to rejecting non-essential cookies. In contrast, the U.S. lacks a comprehensive federal privacy framework. While state-level policies such as the California Consumer Privacy Act (CCPA) emphasize the right to opt out and support browser-based mechanisms like GPC,[8] enforcement is limited and firms retain substantial discretion over both interface design and default settings. As a result, U.S. consent interfaces feature obstructive designs and defaults that favor acceptance more often than their EU counterparts.[9]

2.2. Consent Interfaces

We use Cookie Manager, a browser extension based on the Webmunk framework for browsing-based experiments (Farronato, Fradkin and Karr 2024). Participants install the extension on their Chrome browser. The extension randomizes choice architecture by displaying different consent interfaces that prompt users to make consequential cookie tracking choices.[10]

[7] Prior work also documents persuasion tactics that frame data sharing in a more favorable light, though we do not study these directly.

[8] https://oag.ca.gov/privacy/ccpa/gpc.

[9] https://developers.google.com/tag-platform/security/guides/consent?consentmode=advanced.

[10] For additional details on the extension, see Farronato, Fradkin and Karr (2024). The extension runs exclusively on the desktop Chrome, so the study excludes users of other browsers and mobile devices. Participants are asked to install the extension on their primary device, clear existing cookies upon installation, and all measurements and interventions occur only on that device.

We focus on obstruction and visual manipulation, excluding persuasive language due to its high dimensionality and the resulting need for more treatment arms than our study allows. Figure 1 displays our six consent interfaces. We treat Design C (*Set-Acc-Rej*) as the *baseline interface*. The other interfaces implement obstruction and visual manipulation. For example, Design A (*Acc-Set*) hides the reject option, making rejection only possible through "Cookie Settings" (*deliberate obstruction*); Design D (*Acc-Rej-Set*) places the accept option on top (*reordering options*); and Design F (*Acc-GrayRej-GraySet*) emphasizes the accept button with a brighter color while graying out the other options (*differential salience*). For deliberate obstruction and reordering options, we also include designs that nudge users toward privacy-friendly options (Design B and Design E).

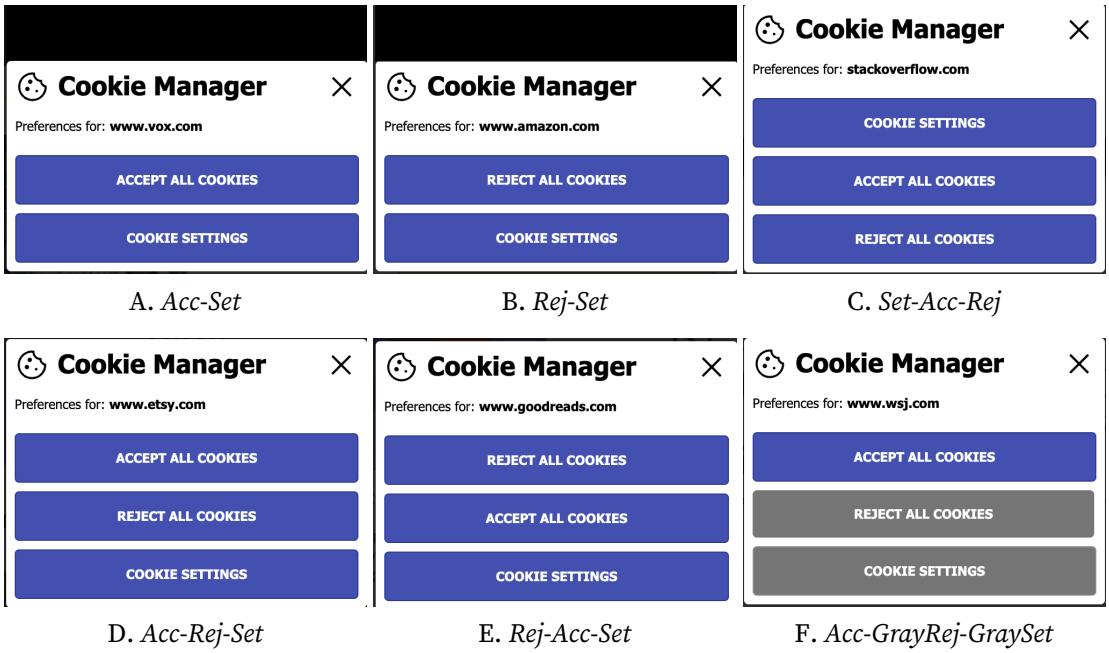
Users can select any displayed option, or close the banner without making an explicit choice by clicking the "*X*" in the top right corner of the banner. When users click on "Cookie Settings," they are presented with six cookie categories to choose from, including information storage & access, performance & analytics, and ad selection delivery & reporting (see the full list in Appendix Table A.1). Selecting all options is equivalent to accepting all cookies; selecting none is equivalent to rejecting all (non-essential) cookies. To minimize choice friction under "Cookie Settings," we allow consumers either to accept all cookies with only one additional click or to reject all cookies with similar ease (as the default on this page is selecting none of the category-specific cookies).

Our banner can appear on any website, regardless of whether the site asks for consent. When an organic cookie choice interface is present, our banner replaces it.[11] To ensure that data-sharing choices are incentive-compatible, Cookie Manager attempts to enforce participants' decisions by using detection rules adapted from open-source packages (e.g., DuckDuckGo's AutoConsent) together with custom scripts to automatically select the corresponding option on each website's native consent interface. While these tools do not achieve perfect enforcement due to factors such as failure to load consent forms or the use of non-standard interfaces, they represent the current state of the art in automated consent management. Importantly, incentive compatibility in our setting does not require perfect enforcement: as long as participants expect their choices to affect tracking with positive probability, they have incentives to reveal their true preferences. When participants close the consent banner without making an active choice, Cookie Manager does not modify any website settings. We explain this enforcement mechanism to participants during onboarding.[12]

[11] Our participants are U.S. residents, so most of them do not see consent banners as often as EU consumers do, due to the lack of federal privacy regulations requiring consent. We did not systematically track how often organic consent banners appeared, nor their designs; instead, we rely on prior literature documenting the prevalence and frequency of manipulative design practices (Utz et al. 2019).

[12] We tell users "After installing the extension, you will see a consent-request popup window whenever you visit a website for the first time. If you make a choice, the extension will try to pass on your choices to the

FIGURE 1. Consent Interface Design across Treatment Groups



Notes: The figure provides screenshots of the six cookie preference interfaces. Captions correspond to the labels used throughout the paper to refer to the treatment conditions. *Acc-Set*: accept-settings; *Rej-Set*: reject-settings; *Set-Acc-Rej*: settings-accept-reject; *Acc-Rej-Set*: accept-reject-settings; *Rej-Acc-Set*: reject-accept-settings; *Acc-GraRej-GraySet*: accept-gray reject-gray settings.

2.3. Study Phases and Randomization

Participants install Cookie Manager in their Chrome browser and face two study phases: a structured survey phase followed by an organic browsing phase.

In the survey phase, participants are asked to visit 20 pre-selected websites spanning different categories and popularity levels (see Appendix A.1 for the full list), presented in randomized order. As participants browse each site, the extension displays a randomly assigned cookie consent banner and records users' choices. This phase serves two purposes. First, it ensures sufficient observations per web domain to estimate choice architecture effects using domain fixed-effect specifications. Second, it complements the organic phase data by providing consent choices when website visits are exogenously given.

In the organic phase, the extension remains active for one week while participants organically browse the web. During this period, the extension shows consent banners on domains where participants have not already interacted with the experimental consent

website. In most cases, if the website has already been collecting consent from users, it will recognize your choice and decide whether to continue tracking you based on your choice."

banners. To reduce disruption to browsing, banners appear at most every 10 minutes, rather than on all domains that participants visit during this phase.[13] At the end of the week, participants complete an exit survey in which we elicit, among other things, their beliefs about default settings. They then uninstall the extension, and each receives $7.50 upon study completion.

Consent interface randomization occurs at the user-by-domain level. In pilot testing, we found no evidence of carryover effects from prior exposures to specific banner designs, so we chose this approach to increase statistical power.

2.4. Framed Field Experiment

Our study is a framed field experiment in the sense of Harrison and List (2004): participants are aware that their browsing behavior is being observed as part of a research study. Two features of this design warrant discussion. First, participant awareness may influence behavior by increasing attention to consent prompts relative to typical browsing or by inducing more deliberate decision-making. To partially mitigate this concern, our design includes both a structured survey phase and an organic browsing phase, the latter capturing decisions made during participants' regular internet use over the course of a week.

Second, we standardize the consent interface across websites. This approach enhances identification by holding design features constant, but it comes at the cost of reduced ecological variation. In particular, users may recognize the study-generated banner and respond differently than they would to heterogeneous, site-specific designs encountered in the wild.

Despite these limitations, our approach represents an important step toward more naturalistic behavior relative to laboratory studies, while preserving experimental control. Moreover, framed field experiments may be among the few feasible approaches available to policymakers, as opposed to individual firms, to compare the welfare implications of alternative consent regimes in a controlled yet behaviorally meaningful setting.

[13] In this second phase, we also randomize users into two frequency treatments: banners appearing at most every 10 minutes (frequent banners) or 60 minutes (infrequent banners). To implement this, a countdown starts after each banner interaction, and a new banner appears after the 10- or 60-minute threshold is crossed and a new domain is visited. We find no significant effect of banner frequency treatment on consent choices (see Appendix E). Because banner exposure is throttled rather than shown on every site, more frequently visited domains are more likely to be treated early on. At the same time, treated domains become ineligible, reallocating subsequent exposure toward infrequently visited sites. In practice, the organic sample includes many less-popular domains. We conduct heterogeneity analyses by domain popularity to account for potential site selection into experiment.

2.5. Sample Description

We recruited participants on Prolific,[14] restricting the sample to U.S.-based, English-speaking adults who use Google Chrome as their primary browser. We pre-registered a target of 800 participants, expecting 640 in the final sample.[15] The realized sample is close to these targets. A total of 1,227 users started the study, 877 consented, and the main sample includes 563 participants who completed the baseline survey and generated valid organic browsing data. We exclude participants who reported making choices based on study expectations rather than their true preferences.[16]

Attrition arises from initial screening, incomplete participation or insufficient interaction data, and restricting the sample to users with at least one cookie selection during the organic phase (Appendix Table A.2). Overall, 61% of eligible participants enter the main sample (53% complete the fully study), which is comparable to similar online experiments (Allcott et al. 2025). Despite attrition, demographics remain broadly stable (Appendix Table A.2), with only minor shifts in gender, education, and income. Other characteristics evolve modestly: retained participants browse more frequently and are slightly more likely to make active choices rather than closing consent banners, whereas familiarity with assigned websites remains similar (Appendix Table A.3).

Table 1 presents descriptive statistics for the main sample. The sample skews slightly male (55%), with an average age of 38. The median household income falls in the $50,000-$74,999 bucket, with 11% of households with an annual income above $150,000. Participants visited an average of 53 unique domains in the week before the study and 54 during the study week (excluding survey-assigned sites). This similarity suggests minimal behavioral response to study participation.[17] Our consent interfaces appeared on 41.2% of the organically visited domains.

We note that participants are self-selected into Prolific and into our study. For this reason, we assess representativeness by comparing our sample to the 2024 American Community Survey.[18] Our sample is a little younger (36 vs. 39 median age), more likely to be male (55% vs. 49%), more educated (52% vs. 36.9% with a bachelor's degree), and lower income (44% vs. 54% above $75k). These differences are typical of online panels.

[14] https://www.prolific.com/. Although Prolific offers a representative sampling option, we did not use it because eligibility and attrition would have affected the representativeness of the final sample.

[15] https://www.socialscienceregistry.org/trials/12862.

[16] We also exclude 3% of users who, instead of the intended lower frequency, saw banners on every new domain during the organic phase.

[17] This information is calculated from participants' Chrome browsing history, which the browser extension collects upon participant consent.

[18] https://data.census.gov/table?y=2024.

TABLE 1. Summary Statistics

		Mean	Median	Std. Dev.
During Survey	Unique Domains in Prior Week	53.27	49.00	39.69
	Domains w. Banner	19.67	20.00	0.82
Post-Survey	Domains w. Banner	22.40	15.00	21.94
	Unique Domains Visited	54.45	39.00	51.19
	Unique URLs	670.75	393.00	817.41
	End Survey Completed	0.86	1.00	0.35
Demographics	Age	37.97	36.00	12.82
	Female	0.45	0.00	0.50
	Bachelor's or Above	0.52	1.00	0.50
	Income > $75,000	0.44	0.00	0.50
Cookie Behavior	Accept-All Rate	0.54	0.64	0.36
	Close-Window Rate	0.26	0.14	0.32
	Reject-All Rate	0.16	0.00	0.28

Notes: The table shows user-level descriptive statistics for the main study sample. Number of participants: 563.

We verify effective randomization of consent banners in two ways. First, we run a proportion test on the distribution of banner designs per website. The proportion test across the survey websites has a p-value of 0.96, which fails to reject the null hypothesis of balanced proportions across treatments. Second, we perform covariate balance tests by regressing user- and domain-level covariates on treatment conditions (Appendix Table B.1). We find no statistically significant difference across designs.

3. Experimental Results

This section presents reduced-form evidence on how consent design affects user behavior. We organize the results around three key mechanisms: 1) visual nudges, 2) defaults and user beliefs, and 3) time costs.

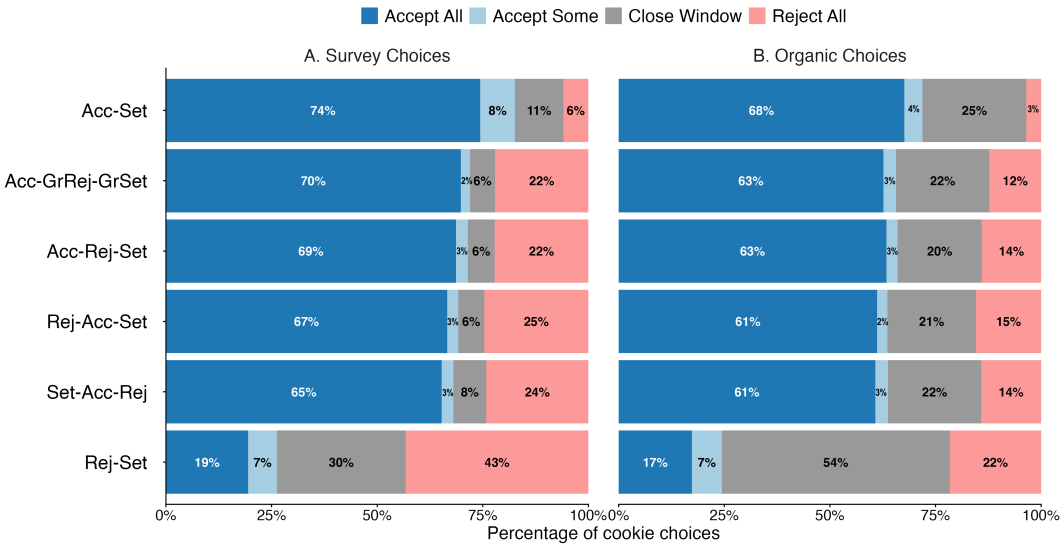
3.1. Baseline Behavior and the Role of Visual Nudges

We start by presenting baseline user behavior. Figure 2 presents the distribution of cookie choices across treatment conditions, separately for the survey (Panel *A*) and organic phases (Panel *B*). Absent deliberate nudging (the *Set-Acc-Rej* row in the figure), participants share their data with websites more than half of the time, with accept-all rates of 65% during the survey phase and 61% during the organic phase.[19] Granular choices, defined as accepting

[19] Our acceptance rates are high but lower than prior evidence in lab studies that also examine privacy choices absent nudging designs, which document the probability of choosing "accept all" to be 77%

only specific cookie categories, occur in only 3% of cases in the baseline condition. Among those making selective choices, 83% accept cookies for *preferences and functionality*, while only 7% consent to *ad selection, delivery, and reporting* (see Appendix Table C.1). Rejecting cookies is substantially less common than accepting, at 24% in the survey phase and 14% in the organic phase. This difference in rejection rates is largely accounted for by an increase in passive behavior: during organic browsing, 22% of users close the consent window without making an explicit choice, up from 8% in the survey phase.

FIGURE 2. Cookie Choices by Experimental Condition



Notes: This figure displays the proportions of cookie choices across banner design treatments. "Accept all" includes instances where a participant clicks into the "settings" page and manually selects all cookies. "Reject all" is similarly defined. Each row corresponds to a treatment condition. The mapping of the labels to each interface is presented in Figure 1.

To quantify the causal effects of banner design on consumer choice, we estimate the following regressions:[20]

$$y_{ij} = \beta_{acc-set_{ij}} + \beta_{acc-grayrej-grayset_{ij}} + \beta_{acc-rej-set_{ij}} + \beta_{rej-acc-set_{ij}} + \beta_{rej-set_{ij}} + \mu_i + \nu_{c(j)} + \epsilon_{ij}. \tag{1}$$

(Vásquez Duque 2024) to 83% (Bielova et al. 2024). Existing reports on ATT consent rates are often lower, at around 44% as of 2024 (see: https://www.appsflyer.com/company/newsroom/pr/att-data-findings/, accessed November 2025). However, Apple uses wording that discourages sharing (Baviskar et al. 2024), and we suspect that some mobile data (such as location) can be more sensitive than web behavior data.

[20] Our treatments often combine different types of manipulations into a single interface. For example, *Acc-Set* simultaneously prioritizes "accept all" and hides "reject all." Appendix D shows the separate effects of individual choice architecture (re-ordering, obstruction, and highlighting) on consumer choices.

Here, i denotes the participant and j denotes the website. We include participant fixed effects μ_i and website category fixed effects $\nu_{c(j)}$, where web categories are obtained via ChatGPT 4o and manual validation.[21] Each β coefficient measures the effect of a specific treatment condition relative to the baseline interface (*Set-Acc-Rej*). We focus on three main outcomes: accepting all, rejecting all, and closing the window without choosing, and cluster standard errors at the participant level.

TABLE 2. Cookie Choices by Experimental Condition

	Accept All (1)	Reject All (2)	Close Window (3)
Acc-Set	0.066***	-0.129***	0.026**
	(0.010)	(0.011)	(0.009)
Acc-GrRej-GrSet	0.030***	-0.015*	-0.015*
	(0.008)	(0.007)	(0.007)
Acc-Rej-Set	0.029***	-0.002	-0.029***
	(0.008)	(0.006)	(0.006)
Rej-Acc-Set	0.005	0.020**	-0.022**
	(0.008)	(0.007)	(0.007)
Rej-Set	-0.444***	0.138***	0.267***
	(0.018)	(0.012)	(0.017)
Benchmark group mean	0.63	0.19	0.16
R^2	0.517	0.469	0.421
Observations	23,685	23,685	23,685
Participant fixed effects	✓	✓	✓
Domain Cat. fixed effects	✓	✓	✓

Notes: Regression results of Equation 1 for three outcomes: accept all cookies, reject all cookies, and close the window without making a choice. The results combine survey and organic choices. Separate estimates for the two sets are presented in Appendix Table C.3. For survey choices, Appendix Table C.4 presents estimates with website fixed effects, which is our pre-registered specification. Standard errors clustered at the participant level. *$p < 0.05$, ** $p < 0.01$; *** $p < 0.001$.

Table 2 reports our main results. Treatments that introduce deliberate obstruction are by far the most effective at steering consumer choices.[22] Hiding the "reject all" option (the *Acc-Set* treatment) reduces rejection rates by 12.9 percentage points (a 68% decline). Symmetrically, hiding "accept all" (the *Rej-Set* treatment) leads to a large reduction in acceptance, lowering acceptance rates by 44.4 percentage points (70.5%). Consumers

[21] We use ChatGPT 4o to classify the websites via the following prompt: "Classify the website domains listed below into one of the following major categories (and only one of the following, do not include categories not in this list and try to limit how often other is selected): 'Reference Website', 'Entertainment Website', 'Business Website', 'E Commerce Website', 'Adult Website', 'News and Portals Website', 'Recreation Website', 'Banking Website', 'Government Website', 'Political Website', 'Other'. Each domain is separated by '+ – – – +'. Please always return a 10-element sequence of classifications separated by '– – –'. The list of domains is: [the list of domains follows]." We then had a research assistant validate the classification through visual inspection.

[22] Our treatments sometimes combine multiple choice architecture elements, so regression 1 does not fully isolate their individual effects. Appendix Table D.1 addresses this by separately coding each element and shows that the effects in our main analysis are driven primarily by obstruction.

substitute toward the remaining options. When "reject all" is hidden, users are more likely to accept all cookies (an increase of 6.6 percentage points). In contrast, when "accept all" is hidden, users primarily substitute toward closing the window (up 26.7 percentage points), and to a lesser extent toward rejecting cookies (up 13.8 percentage points). Finally, hiding either option from the main interface increases the likelihood that users access the settings menu and make granular choices. However, this effect remains quantitatively small: even under these treatments, granular choices rise only modestly and reach at most 7 or 8%.

In contrast, purely visual manipulations have modest effects. Placing "accept all" at the top (*Acc-Rej-Set*) increases acceptance by 2.9 percentage points (a 4.6% increase), whereas placing "reject all" at the top (*Rej-Acc-Set*) leads to a small increase in rejections of 2 percentage points (10.5%). Highlighting "accept all" and placing it at the top (*Acc-GrRej-GrSet*) yields a similarly modest increase in acceptance of 3 percentage points, an effect that is statistically indistinguishable from simple reordering without additional visual emphasis.[23]

3.2. Heterogeneity in Consent Behavior

Figure 3 shows variation in acceptance rates across website categories, estimated as fixed effects in Equation 1. Users are less likely to accept cookies on more sensitive domains (e.g., adult, political, and government websites) than for more mundane browsing (entertainment, e-commerce, and banking sites). Familiarity and popularity also matter: as shown in Panels *a* and *b* of Table 3, participants are more likely to accept cookies on websites they have previously visited or recognize, and on more popular domains. We find limited heterogeneity in the effects of visual nudges across these dimensions (Appendix Tables C.5a and C.5b).

Turning to user-level heterogeneity, observable demographics have limited explanatory power. Women are somewhat less likely to accept cookies, and more frequent internet users are more likely to close the banner. Stated privacy attitudes are only weakly correlated with observed behavior (Panel *d* of Table 3).

To assess the relative importance of user- versus website-level variation, we estimate a random effects model (Appendix Table A.4), which shows that dispersion in user-specific effects is more than five times larger than that of website effects.

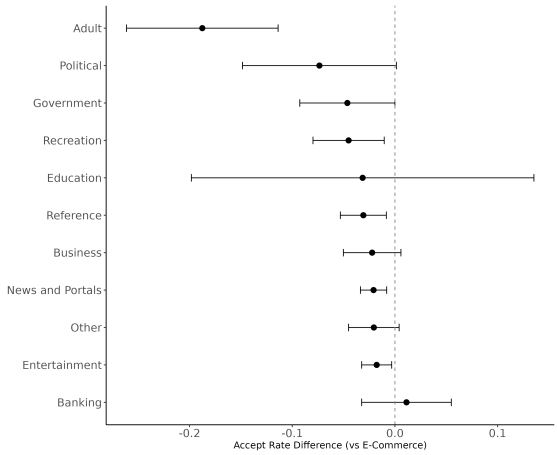
[23]The limited impact of visual manipulations, relative to deliberate obstruction, is consistent with prior evidence from more controlled or survey-based settings. For example, Habib et al. (2022) compare a design where the reject option is hidden to one where rejecting is the default and find large differences in behavior. In addition, Utz et al. (2019) and Vásquez Duque (2024) find minimal effects from salience-based interventions.

TABLE 3. Heterogeneity in Cookie Choices Across Websites and Users

	Accept All (1)	Reject All (2)	Close Window (3)
Panel a: Experience based on survey answers (survey data only)			
Normally Visit	0.071***	-0.072***	-0.010
	(0.009)	(0.010)	(0.006)
Heard Of	0.026**	-0.026**	-0.002
	(0.009)	(0.009)	(0.006)
R^2	0.652	0.587	0.562
Observations	11,075	11,075	11,075
Panel b: Experience based on browsing history and site popularity			
Pre-Exp Visit	0.018*	-0.031***	0.011
	(0.008)	(0.006)	(0.006)
Domain Rank (Log 10)	-0.010***	0.002	0.007***
	(0.002)	(0.002)	(0.002)
R^2	0.518	0.470	0.422
Observations	23,685	23,685	23,685
Panel c: User characteristics			
Bachelor's or Above	-0.086	0.017	0.056
	(0.044)	(0.034)	(0.039)
Income > $75,000	-0.004	-0.001	0.001
	(0.031)	(0.023)	(0.023)
Age	0.001	-0.001	-0.002**
	(0.001)	(0.001)	(0.001)
Female	-0.083**	0.048*	0.042
	(0.029)	(0.023)	(0.023)
Prior Domains Visits (Log)	-0.012	-0.013	0.019*
	(0.013)	(0.011)	(0.009)
R^2	0.147	0.069	0.122
Observations	22,460	22,460	22,460
Panel d: Attitudes about privacy			
Privacy: Meaningful Decision	0.025	-0.005	-0.030
	(0.035)	(0.028)	(0.024)
High Value Privacy	-0.013	0.070*	-0.078
	(0.049)	(0.028)	(0.043)
R^2	0.135	0.063	0.113
Observations	23,685	23,685	23,685

Notes: Regression results of Equation 1 augmented with measures of website familiarity, popularity, and user characteristics. In Panel *a*, we add two dummies to indicate whether the study participant has heard of the website and whether the study participant normally visits the website (both questions are answered as part of the intake survey, so we only include consent choices during the survey phase). In Panel *b*, we aggregate across browsing and survey data, allowing for a level shift between the two. The dimensions of heterogeneity are proxied by a dummy indicating whether the study participant visited the website in the two weeks preceding the study (obtained from their Chrome browsing history) and by the website's popularity rank (from Tranco logs, https://tranco-list.eu/). Panel *c* includes user characteristics (age, gender, income, education, and the number of prior domain visits from their browsing history). Panel *d* focuses on self-reported attitudes towards privacy. Regressions in Panels *c* and *d* require us to exclude individual fixed effects. $^{*}p < 0.05$, ** $p < 0.01$; *** $p < 0.001$. $^{*}p < 0.05$, ** $p < 0.01$; *** $p < 0.001$.

FIGURE 3. Consent Rates by Site Category



Notes: Differences in cookie acceptance rates by website category. The plot shows the estimated category fixed effects from Equation 1, where the outcome is "accept all" and the baseline category is e-commerce websites. Bars denote 95% confidence intervals.

Motivated by this substantial user-level heterogeneity, we examine differences in privacy preferences in the absence of deliberate nudging. To this end, we partition participants into three groups based on their choices in the baseline interface: *Always Accept*, who consistently accept all cookies; *Never Accept*, who consistently reject cookies; and *Some Accept*, whose choices vary across websites within the baseline condition.[24] Table C.2 shows that these groups mostly differ in their education level and internet browsing intensity, with the *Some Accept* group being more active online and the *Never Accept* group being more educated. These patterns motivate our subsequent latent class model specification.

3.3. Defaults and User Beliefs

A central feature of the data is the prevalence of passive behavior. While participants respond similarly to visual nudges across the survey and organic phases, Figure 2 shows that they are substantially more likely to close the consent banner without making an explicit choice during organic browsing. As shown in Table 2, column (3), closing the window is also an important margin of substitution in response to obstruction: when preferred options are hidden, users frequently disengage rather than actively switching to an alternative.

[24]The categorization of these groups excludes passive choices. For example, a participant who accepts all cookies for 90% of the baseline interfaces and closes consent windows for the remaining 10% would still belong to the "Always Accept" group.

Despite the frequency of closing the banner, participants hold heterogeneous beliefs about default settings. Among those who fill out our endline survey, 61% of respondents report believing that closing the banner results in rejecting cookies, while 25% believe it leads to acceptance, and the remaining 14% are uncertain. Given the absence of comprehensive federal regulation and the resulting incentives for websites to default to tracking in the U.S., most users misperceive the consequences of dismissing the banner.

Accounting for users' beliefs yields similar implied preferences across the survey and organic phases. Indeed, imputing passive choices using belief data, we estimate that 63% of users in the survey phase and 62% in the organic phase intend to accept cookies, while 30% in both phases intend to reject them.

In the endline survey, we asked participants their reasons for accepting or rejecting cookies and their broader views on data sharing. The reported motivations suggest that observed choices reflect considered preferences. Participants response show that acceptance decisions are often motivated by trust (38.6%: "it was a website I totally trusted"), convenience (21.9%: "it was just easier to do so"), and functionality (17.6%: "to make the websites function well"). Rejection decisions reflect parallel considerations: distrust of unfamiliar websites (26.2%: "did not trust site"), concerns about tracking or advertising (7.4%: "I didn't want to be tracked by advertisers on that website"), and general privacy preferences (13.2%: "in some website I felt not to accept some cookies because I may not want the website to have my data").

More broadly, respondents demonstrate a reasonable understanding of the trade-offs associated with data sharing. 70% identify personalized advertising and 15% improved functionality as benefits, while over three quarters cite privacy loss (43%), security risks (18%), or increased advertising (15%) as key costs. Only 4% report being unsure about what cookies do.

3.4. Time Costs of Consent

To examine the time costs of consent, we measure interaction time as the elapsed time from banner appearance to the user's final action, including intermediate clicks and navigation. We censor observations at 60 seconds, well above the 99th percentile, to exclude likely task-switching rather than genuine interaction. We estimate Equation 1 using time spent as the outcome.

Table 4 reports the results. In the baseline design, users spend 5.25 seconds per banner in the survey phase and 7.34 seconds in the organic phase (2.46 seconds at the

TABLE 4. Time Spent by Experimental Condition

	Survey	Organic
	Time Spent (Seconds)	
	(1)	(2)
Acc-Set	0.497	-0.524
	(0.291)	(0.382)
Acc-GrRej-GrSet	-0.144	-0.192
	(0.259)	(0.393)
Acc-Rej-Set	-0.426	0.144
	(0.234)	(0.421)
Rej-Acc-Set	-0.257	-0.363
	(0.242)	(0.379)
Rej-Set	2.125***	1.110**
	(0.281)	(0.423)
Benchmark group mean	5.25	7.34
R^2	0.193	0.120
Observations	11,075	12,565
Participant fixed effects	✓	✓
Domain Cat. fixed effects	✓	✓

Notes: Regression results of Equation 1 where the outcome is time spent interacting with the cookie consent banner. Results are shown separately for survey choices (column 1) and organic choices (column 2). Standard errors are clustered at the participant level. *$p < 0.05$, ** $p < 0.01$; *** $p < 0.001$.

median). The only design that substantially increases interaction time is *Rej-Set*, which hides the most commonly chosen option.[25]

Additional clicks are a primary driver of these costs. Table 5 shows that accessing the settings menu, which is required when preferred options are hidden, adds over 4 seconds on average and 3.38 at the median. This friction amounts to more than half of the total baseline interaction time.

Though modest per banner, these frictions impose nontrivial costs on users. At the average hourly wage of $36,[26] 7.34 seconds per domain across 54 domains per week translates to a weekly time cost of approximately $4 per user ($1.30 using the median time). Time costs will be a central input in the welfare analysis in the following sections.

[25]This additional time cost may also reflect confusion or unfamiliarity, as interfaces that hide "accept all" are uncommon in practice.

[26]This number is based on Bureau of Labor Statistics estimates (https://www.bls.gov/news.release/empsit.t19.htm).

TABLE 5. Extra Time Spent When Clicking on Settings

	Time Spent (Seconds)
	(1)
User Clicks Settings	4.069***
	(0.442)
Close Window	1.825***
	(0.409)
Accept Selected	3.282***
	(0.673)
R^2	0.102
Observations	22,842
Condition fixed effects	✓
Participant fixed effects	✓
Domain Cat. fixed effects	✓

Notes: The table reports regression estimates of time spent as a function of whether the user clicks on settings. The baseline is the time spent on a banner in the baseline condition. The indicator *User Clicks Settings* equals one if the user accesses the settings menu. Standard errors are clustered at the user level. $^{*}p < 0.05$, ** $p < 0.01$; *** $p < 0.001$.

4. Structural Model

To quantify the welfare implications of different consent choice architectures, we develop a structural model of consumer data sharing decisions that incorporates the influence of consent design. The model translates observed choices into welfare measures and quantifies the costs imposed by design frictions. More importantly, it enables us to evaluate policy interventions that vary choice frictions, defaults, and browser-level vs. site-by-site consent.

4.1. Model and Estimation

Given a cookie consent banner, a consumer can choose among four actions: {accept all, reject all, customize settings, close window}. We model consumers' decision utility as a function of the selected option and the frictions introduced by the consent interface design. Here, we use the notion of *decision utility* (Kahneman and Thaler 1991) to acknowledge that consumers may not always choose the option that maximizes their welfare, because they may hold mistaken beliefs and their choices can be affected by salience manipulations. For notational simplicity, we omit consumer-specific subscripts below.

Let θ_k denote the value that the consumer assigns to data-sharing option k. Specifically, θ_{acc} corresponds to the value of accepting all cookies, θ_{rej} to rejecting all cookies (normalized to zero for identification), and θ_{set} to customizing settings.[27]

The perceived value of closing the consent banner depends on a consumer's belief about the website's data collection default. Our endline survey elicits each participant's subjective probability, ρ_{belief}, that closing the window will result in accepting all cookies. Their perceived utility is thus $\rho_{\text{belief}}\theta_{acc}$ (recall $\theta_{rej} = 0$). For respondents who believe that closing the window means accepting cookies (22%), we set $\rho_{\text{belief}} = 1$; for those who believe it means rejecting cookies (52%), we set $\rho_{\text{belief}} = 0$. For participants who report uncertain beliefs (12%) or did not answer the belief question (14%), we estimate their beliefs as ρ_{unc} and ρ_{miss}, respectively. Under the assumption of risk neutrality, ρ_{unc} and ρ_{miss} can be interpreted as the subjective probability these participants assign to full acceptance upon banner closure.[28]

Closing the banner without making an explicit choice is also a distinct action whose attractiveness may vary over time. We parameterize its cost with two terms: a constant κ_1 and a time-dependent term κ_2/t_i, where t_i represents the number of days since user i's enrollment in the study (with the enrollment day counted as day 1). The time-dependent term captures the fact that banner closure increases over the study period but stabilizes after the first few days (Appendix Figure A.2).

Interface design affects decisions through two channels. First, deliberate obstruction increases friction through extra clicks: consumers incur an additive cost C_{set} to access the settings menu. Second, visual manipulations alter salience and thus the decision utility of affected options. We model their effects as multiplicative factors applied to the baseline utility. To accommodate potentially negative utilities θ_k while ensuring proper estimation of the δ parameters, we specify the salience-adjusted utilities as $\theta_k + \delta_{\text{top}}\theta_k^2$ for top placement and $\theta_k + \delta_{\text{gray}}\theta_k^2$ for graying out, where we expect $\delta_{top} \geq 0$ and $\delta_{gray} \leq 0$. This formulation ensures that salience effects scale proportionally with the magnitude of baseline utility, regardless of sign.

For ease of comparison, Table 6 summarizes the complete utility specification across all treatment conditions.

[27] Our design does not separately identify the benefit of granular cookie selection from the hassle cost of doing so. As a result, θ_{set} is the net utility of the customization benefit and the hassle cost of making granular selections.

[28] We do not discount the belief of participants who believe that closing the window results in either full acceptance or full rejection. Our descriptive evidence in Section 3.3 indicates that these users substitute between closing the window and their belief-consistent action at roughly a 1:1 ratio.

TABLE 6. Decision Utility of Each Option Across Treatment Conditions

	Option			
Treatment Design	Accept All	Reject All	Customize Settings	Close Window
Acc-Set	$\theta_{acc} + \delta_{top}\theta_{acc}^2$	$-C_{set}$	$\theta_{set} - C_{set}$	$\rho_{belief}\theta_{acc} + \kappa_1 + \frac{\kappa_2}{t_i}$
Rej-Set	$\theta_{acc} - C_{set}$	0	$\theta_{set} - C_{set}$	$\rho_{belief}\theta_{acc} + \kappa_1 + \frac{\kappa_2}{t_i}$
Set-Acc-Rej	θ_{acc}	0	$\theta_{set} + \delta_{top}\theta_{set}^2 - C_{set}$	$\rho_{belief}\theta_{acc} + \kappa_1 + \frac{\kappa_2}{t_i}$
Acc-Rej-Set	$\theta_{acc} + \delta_{top}\theta_{acc}^2$	0	$\theta_{set} - C_{set}$	$\rho_{belief}\theta_{acc} + \kappa_1 + \frac{\kappa_2}{t_i}$
Rej-Acc-Set	θ_{acc}	0	$\theta_{set} - C_{set}$	$\rho_{belief}\theta_{acc} + \kappa_1 + \frac{\kappa_2}{t_i}$
Acc-GrRej-GrSet	$\theta_{acc} + \delta_{top}\theta_{acc}^2$	0	$\theta_{set} + \delta_{gray}\theta_{set}^2 - C_{set}$	$\rho_{belief}\theta_{acc} + \kappa_1 + \frac{\kappa_2}{t_i}$

Notes: The table reports the decision utility associated with each option across the six treatment conditions. Although not explicitly indicated with subscripts, the parameters ($\theta_{\text{acc}}, \theta_{\text{set}}, C_{\text{set}}, \delta_{\text{top}}, \delta_{\text{gray}}$) are functions of the consumer type ι. In contrast, the costs of closing the window (κ_1, κ_2) and the belief parameters $\rho_{\text{belief}} = (\rho_{\text{unc}}, \rho_{\text{mis}})$ are assumed to be constant across consumer types.

A consumer i picks the option k that maximizes their utility given the banner design on website j:

$$\underset{k\in\{acc,rej,set,close\}}{Max} U_{ijk|design_{ij}},$$

where $U_{ijk|design_{ij}} = u_{ik|design_{ij}} + \epsilon_{ijk}$. The first component is the user i's decision utility from option k (which depends on banner design as presented in Table 6), whereas the random component ϵ_{ijk} follows an i.i.d. Type-1 extreme value distribution.

To capture preference heterogeneity, we estimate a latent class model (Kamakura and Russell 1989) that allows for three latent consumer types ι with shares below:[29]

$$s_{i\iota} = \frac{exp(\lambda_{i\iota})}{\sum_{\iota'\in\{1,2,3\}} exp(\lambda_{i\iota'})}. \tag{2}$$

We let the shares depend on the user's education and browsing activity in the week preceding the study: $\lambda_{i\iota} = \lambda_\iota^1 + \lambda_\iota^2 \mathbf{1}\{\text{education}_i \geq \text{bachelor degree}\} + \lambda_\iota^3 log(1+\text{prior domain visits}_i)$, with normalization $\lambda_3 = 0$. We include education and prior browsing intensity in the first stage because our descriptive evidence suggests that these demographics are the most predictive of heterogeneous baseline consent choices (Appendix Table C.2).[30]

Each consumer type ι has distinct preferences and choice architecture sensitivity parameters $(\theta_{acc}, \theta_{set}, C_{set}, \delta_{top}, \delta_{gray})_\iota$, while belief parameters (ρ_{unc}, ρ_{mis}) and closing

[29] As a robustness check, we also estimate models with 2 and 4 classes, both of which yield worse fit according to the Bayesian Information Criterion. In particular, the 4-class model splits either the discerner or the rejector group, with coefficients similar between the new group and one of the existing groups. The estimates for these alternative models are available in Appendix Tables F.1 and F.2.

[30] Alternative specifications, such as including income and education levels, generate a worse fit without meaningfully changing the key parameter estimates.

window costs (κ_1, κ_2) are common across types. Conditional on a user being type ι, the probability of choosing option k is

$$P_{\iota jk} = \frac{exp(u_{\iota k|design_{ij}})}{\sum_{k\in\{acc,rej,set,close\}} exp(u_{\iota k|design_{ij}})}. \tag{3}$$

The unconditional probability of selecting option k by consumer i is thus the product of a consumer's probability of being a certain type and the type-specific choice probability:

$$P_{ijk} = \sum_{\iota\in\{1,2,3\}} s_{i\iota} P_{\iota jk}. \tag{4}$$

Identification of the model parameters relies on our experimental variation. Consumer preferences for specific data sharing options are identified from choices under the baseline treatment without deliberate nudging. Friction costs are identified by comparing choice probabilities when options are hidden versus visible on the main screen. Salience effects are identified by comparing choices between interfaces that reorder or gray out options while maintaining the same choice set. Beliefs about what happens upon banner closure are elicited through the endline survey, which helps us infer the potential mistakes and costs associated with closing the banner. Multiple observations of the same consumer over time help identify how the utility of closing the window varies over the duration of the study.

We use Equation 4 to form our likelihood function, and estimate the model parameters by maximum likelihood using data from both the survey and organic phases. We cluster standard errors at the participant level.

4.2. Results

Panel A of Table 7 presents the estimates for consumer utility parameters. Type-1 users have the highest utility from accepting all cookies ($\theta_{acc} = 2.66$), so we label them "acceptors." Type-2 users strongly dislike data sharing ($\theta_{acc} = -4.96$), so we label them "rejectors." Type-3 users, labeled "discerners," have a moderate utility for accepting all cookies ($\theta_{acc} = 1.21$) and a greater willingness to customize settings compared to the other two groups ($\theta_{set} = 0.62$).[31]

The friction parameters confirm that interface designs impose substantial behavioral costs. The cost of clicking the "settings" button (C_{set}) ranges from 1.7 to 2.9 across con-

[31]The estimated preference parameters naturally give rise to three groups that we label "acceptors," "rejectors," and "discerners." These labels are shorthand for dominant tendencies rather than deterministic behavior.

TABLE 7. Parameter Estimates Across Consumer Subsets

Parameter	Explanation	Estimates		
Panel A: Utility Parameters				
Type-Specific Parameters		Type 1 "Acceptors"	Type 2 "Rejectors"	Type 3 "Discerners"
θ_{acc}	Utility of accepting all	2.660***	-4.960***	1.210***
		(0.157)	(0.736)	(0.085)
θ_{set}	Utility of granular choice	0.407	-0.280*	0.615***
		(0.355)	(0.118)	(0.158)
C_{set}	Cost of clicking settings	2.900***	2.650***	1.660***
		(0.200)	(0.166)	(0.116)
δ_{top}	Effect of ranking on top	0.422***	0.029	0.076
		(0.082)	(0.042)	(0.077)
δ_{gray}	Effect of being grayed out	-1.610	-4.410	-0.041
		(10.200)	(5.030)	(0.412)
Pooled Parameters				
κ_1	Utility of closing window		0.752***	
			(0.045)	
κ_2	Inclination of closing over time		-1.670***	
			(0.070)	
ρ_{unc}	Pr(accept | close) for uncertain		0.352***	
			(0.025)	
ρ_{miss}	Pr(accept | close) for missing		0.469***	
			(0.138)	
Panel B: Latent Class Probability Parameters				
λ^1	Intercept	0.573***	-0.510**	
		(0.165)	(0.194)	
λ^2	Bachelor degree or higher	-1.720***	-0.874**	
		(0.426)	(0.305)	
λ^3	Nr. domains in browsing history	-1.470***	-1.520***	
		(0.238)	(0.214)	
Share of Users		0.454	0.176	0.369
Log likelihood		-23,019.78		
Bayesian information criterion		46,291.38		

Notes: The table presents parameter estimates from the utility model. Panel A reports estimates for consumer utility parameters, while Panel B reports estimates for the latent model parameters. Given the parameter estimates from Panel A, we refer to Type 1 users as "acceptors," Type 2 users as "rejectors," and Type 3 users as "discerners." Alternative estimates with 2-type and 4-type latent models are presented in Appendix tables F.1 and F.2. Validation of predicted choices compared to actual choices is presented in Appendix Table A.3. Standard errors are clustered at the participant level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

sumer types and exceeds the utility difference between accepting and rejecting cookies for acceptors and discerners. In comparison, visual salience effects are modest and statistically insignificant for most types. Overall, these results align with our reduced-form findings that obstruction effects are much larger than those from visual manipulation.

The cost of closing the banner without making a choice ($\kappa_1 + \frac{\kappa_2}{t_i}$) is initially high relative to rejecting cookies, estimated as -0.92 on the first day of the study. However, the closing option becomes increasingly attractive over time as the time-dependent component $\frac{\kappa_2}{t_i}$ increases. By day 3, closing the banner becomes more attractive than explicitly rejecting cookies. Finally, the belief parameters (ρ) suggest that respondents who are uncertain about the website default assign a 35% probability that closing the window implies accepting cookies. For participants who did not complete the endline survey, this probability is 47%.

Panel *B* of Table 7 shows how consumer types vary across the population. Acceptors form the largest group at 45% of users, followed by discerners at 37% and rejectors at 18%. Both education and prior browsing intensity significantly predict consumer type. College-educated users are the least likely to be acceptors and most likely to be discerners. Similarly, users with more extensive browsing histories are significantly less likely to be classified as acceptors or rejectors than as discerners.[32]

5. Consumer Surplus

The structural model estimates enable us to assess the welfare impacts of existing and proposed consent policies. These policies vary along three dimensions: permitted banner designs, defaults upon inaction, and the frequency of user decisions. A common approach mandates interfaces without obstruction while retaining site-by-site consent.[33] Under site-specific consent, defaults are critical because many users close banners without making a choice; we therefore compare pro-privacy and pro-data-sharing defaults. An alternative is browser-level consent, or GPC,[34] which shifts decisions from each website to the browser. GPC reduces interaction costs but limits customization, making its welfare effects ex ante ambiguous.

We thus evaluate consumer welfare under the following policies:

[32] As a robustness check, we also estimate models with 2 and 4 latent classes, both of which have worse out-of-sample fit according to the Bayesian Information Criterion. See Appendix Tables F.1 and F.2 for details.

[33] See German court ruling requiring a visible and equally salient "reject all" button (https://cybernews.com/privacy/reject-all-cookies-button-must-be-present-in-europe/), and California Privacy Protection Agency guidance mandating "symmetry in choice" (https://www.dwt.com/blogs/privacy--security-law-blog/2024/09/california-guidance-on-dark-patterns-and-privacy).

[34] See https://globalprivacycontrol.org/.

1. **U.S. status quo**: An *Acc-Set* design (i.e., a design that uses deliberate obstruction and reordering), with websites defaulting to collect all cookies when the user does not make an explicit choice;
2. **EU norm**: An interface without deliberate nudging (*Set-Acc-Rej*), with websites defaulting to no cookie tracking upon inaction;
3. **Optimal Banner Interface**: An interface without nudging (*Set-Acc-Rej*), with the default set to the majority preference, i.e., collect all cookies upon inaction;
4. **Optimal Banner Interface with Correct Beliefs**: An interface without nudging (*Set-Acc-Rej*), with the default set to collect all cookies upon inaction, and with consumers having correct beliefs about the default settings;[35]
5. **Global Accept**: A browser-level configuration that automatically accepts all cookies across websites;
6. **Global Reject**: A browser-level configuration that automatically rejects all cookies across websites;
7. **Global Privacy Control**: A browser-level configuration that lets users choose one of three global options for all websites: accept all cookies, reject all cookies, or apply a uniform customized preference.

We assign different defaults to the U.S. and EU scenarios to match prevailing industry practices (see Section 2). Major consent frameworks (e.g., Google Consent Mode) tailor defaults to regional regulation: under GDPR, tracking is typically set to "denied" until explicit consent is given, whereas in the U.S. defaults are generally more permissive in the absence of binding federal rules. Accordingly, we assume that inaction implies rejection under the EU norm and acceptance under the U.S. status quo. The default under the optimal banner design is set to acceptance, reflecting the majority preference in our sample.

For welfare, we first need to distinguish between decision utility, which governs observed choices and may be distorted by beliefs or interface design, and welfare utility, which reflects true consumer surplus. Second, we need to incorporate the time cost of interacting with consent banners. Finally, we want to express welfare in dollar terms using the opportunity cost of time. We discuss each of these challenges in turn.

Following Kahneman and Thaler (1991), we treat preferences in the *baseline interface* as welfare-relevant. Choice architecture can distort decision utility through incorrect beliefs about defaults or visual salience, without affecting underlying welfare utility. For site-by-site consent policies, we therefore allow choices to depend on decision utility, but evaluate surplus using welfare-relevant utility, following Train (2015). Consumer i's

[35] Due to the absence of belief mistakes and salience distortion, consumers' utility in this state would be the "welfare relevant" utility per Kahneman and Thaler (1991).

surplus is:

$$CS_i(\text{banner-level policy}) = \sum_{\iota} s_{i\iota} \left[\log \left(\sum_k e^{u_{ik\iota|policy}} \right) + \sum_k P_{ik\iota|policy} \cdot d_{ik\iota|policy} \right],$$

where the first term is the standard logit surplus based on decision utility and the second term is a correction for differences between expected decision utility and expected welfare utility. In the second term, $P_{ik\iota\,|policy}$ is the choice probability for option k, and $d_{ik\iota\,|policy}$ is the difference between the welfare and the decision utility. The cost of closing the banner declines over time. To capture long-run behavior, we evaluate utility at its steady state (formally, $\lim_{t\to\infty} \kappa_1 + \kappa_2/t = \kappa_1$).

Beyond nudges, consent banners impose time costs that directly reduce welfare. Browser-level policies eliminate repeated interactions but restrict site-specific customization. Under such policies, consumer surplus becomes:

$$CS_i(\text{browser-level policy}) = \sum_{\iota} s_{i\iota} u_{ik^*\iota|policy} + V_{\text{banner time}},$$

where k^* denotes the chosen global setting. The first term captures the utility from choices made without customization, as users select $k^* = \arg\max_k u_{ik\iota}$ based on average utility across sites, forgoing site-specific customization. The term $V_{\text{banner time}}$ captures the benefit of avoiding repeated banner interactions. We assume that the value from time savings is proportional to time spent, $V_{\text{banner time}} = C_{\text{set}} \times$ (time spent per banner/time to click "settings") , and recover $V_{\text{banner time}}$ from the estimated cost of accessing settings.

To express welfare in dollar terms, we map the time cost of accessing settings (C_{set}) into monetary units using participants' opportunity cost of time. We back out the dollar value of C_{set} by multiplying the time spent per interaction by an implied hourly wage based on participant income. In our data, accessing the settings menu takes on average 4 seconds per domain (Table Table 5). Using the midpoint of each participant's reported income bracket as the imputed hourly wage, this implies a per-interaction cost of $(\$42/3600) \times 4$. For a participant in the \$75–100k bracket (hourly wage \$42), this corresponds to a weekly cost of \$2.5 given an average of 54 domains visited. We use this mapping between C_{set} and its dollar value to convert all welfare measures into monetary terms.

Our approach assumes that cost parameters are proportional to time spent and that time is valued at the hourly wage. It abstracts from additional annoyance or cognitive costs generated by consent banners (Bernheim, Kim and Taubinsky (2024)), as well as from differences in how users value browsing versus working time. These omissions likely operate in opposite directions: the former would lead us to understate friction costs, while the latter may overstate the opportunity cost of time. Rather than adjusting for these

effects directly, we later conduct a break-even analysis to quantify the time valuation that would rationalize indifference across site-by-site and browser-level policies.

Table 8 reports our consumer surplus estimates in dollar terms. The first column shows results averaged across types, while the remaining columns highlight heterogeneity across the three consumer types. The per-user-week effects appear modest in absolute terms, but they represent meaningful welfare changes that accumulate over time.

TABLE 8. Consumer Surplus Under Counterfactual Policies ($, Weekly)

Counterfactual	Average	Type 1 Acceptors	Type 2 Rejectors	Type 3 Discerners
U.S. Status Quo	2.36	2.76	-2.47	4.18
EU Norm	2.77	2.67	1.33	3.58
Optimal Banner Interface	2.96	3.26	-0.61	4.31
Optimal Banner Interface (with correct beliefs)	3.39	3.6	0.61	4.46
Global Accept	5.81	6.94	0.37	7.01
Global Reject	4.98	4.76	5.03	5.24
Global Privacy Control	6.63	6.94	5.03	7.01

Notes: The table reports weekly consumer surplus per user in dollar units under various counterfactual policies. The U.S. status quo refers to an *Acc-Set* interface combined with an "accept all" default; the EU norm refers to the baseline interface with a "reject all" default; the optimal banner interface refers to the baseline interface with an "accept all" default; the optimal banner interface with correct beliefs ensures that people closing the window have correct beliefs about default settings. "Global Accept" and "Global Reject" force each individual to either always accept or always reject all cookies. The last row, "Global Privacy Control," allows each individual to make their preferred global choice. Columns 2-4 correspond to the estimates by user type, and the "average" column refers to the estimate averaged across latent types.

We begin by comparing consent-based policies. The optimal banner interface, a design without deliberate nudging with an accept-all default, yields an average consumer surplus of $2.96 per user-week. This policy performs well for acceptors and discerners relative to other consent based policies. Yet, it imposes welfare costs on rejectors because the default does not match their preferred option and they mistakenly believe that closing the window rejects cookies.

The difference between the U.S. status quo and the optimal banner interface reflects the effect of deliberate nudges, especially obstruction. Allowing nudges decreases consumer surplus by $0.6 per user-week (a 20% decline relative to the optimal banner interface). Rejectors lose the most from the U.S. status quo, as the reject option becomes harder to access.

The difference between the EU norm and the optimal banner interface reflects the role of default settings. Changing the default to rejecting cookies (away from the majority preference) decreases consumer surplus by $0.19 per user-week (a 6% decline relative to the optimal banner interface). While acceptors and discerners are hurt, moving to the EU norm comes at a substantial welfare gain for rejectors, because the default aligns with

their preferences. The EU norm still outperforms the U.S. status quo, increasing average surplus by $0.41 per user-week, due to its use of a baseline design that reduces choice friction.

Correcting beliefs about default settings increases consumer surplus by an additional $0.43 per user-week (a 15% gain relative to the optimal banner design). The gains are once again concentrated among rejectors, who previously closed the banner under the mistaken belief that doing so rejected cookie tracking. Once beliefs are corrected, these mistaken choices are eliminated, leading to large welfare improvements for this group.

Remarkably, GPC outperforms site-by-site consent by a large margin. Relative to the optimal banner interface, it increases consumer surplus by $3.67 per user-week, more than doubling surplus. Rejectors benefit the most from GPC, with an increase of $5.64 per week relative to the optimal interface banner. Acceptors gain $3.68, while discerners, despite benefiting the least, still gain $2.70.

Overall, these results highlight three insights. Removing obstruction raises welfare for all users, and GPC strictly dominates site-by-site consent across all consumer types. In contrast, default settings generate distributional trade-offs: users whose preferences align with the default benefit, while others incur losses. As a result, the optimal default depends on the underlying population. In our setting, the optimal default happens to be accept-all for our participants, who may be less privacy-conscious than the broader population.

Our findings are robust and likely to generalize. We first consider an alternative surplus calculation based on median interaction times. At the median, a participant spent 2.46 seconds interacting with the baseline banner and 3.38 seconds accessing settings. Using these values, GPC continues to outperform other designs, increasing consumer surplus by $0.88 per user-week relative to the best site-by-site policy, a 36% gain (Appendix Table A.5). The only exception is that, under these more conservative interaction times, discerners with correct beliefs would weakly prefer the optimal banner interface to GPC by a small margin of $0.06 per week.

Next, we compute the interaction time that would make the optimal banner interface welfare-equivalent to GPC. This break-even occurs at 1.17 seconds per banner, just 16% of the observed average. Equivalently, users would need to value their time at only $6.10 per hour for the two policies to yield equal welfare.

Stated preferences from the endline survey align with our revealed-preference estimates. 62.4% of participants report strictly preferring a GPC tool, compared to 13.4% who prefer site-by-site consent (Appendix Figure A.4). The average stated willingness to pay for such a tool is $4, similar in magnitude to our estimated surplus gains from moving

from the U.S. status quo to GPC. While stated and revealed preferences need not coincide, this alignment provides reassurance that our welfare estimates are well calibrated.

Finally, we assess the external validity of our findings. Although our main sample consists of U.S. consumers, a follow-up survey of EU and U.S. consumers indicates that preferences over consent architectures are broadly similar to those of our study participants. We provide additional details in Appendix G.

6. Discussion

The debate over privacy policy is characterized by competing interests: businesses championing increased data collection versus regulators who often assume consumers prioritize data minimization. We find that consumers' choices to share data are sensitive to how consent is presented. Deliberate obstruction, which makes privacy-preserving options harder to access, is the design pattern with the largest effects among the designs we study. In contrast, visual manipulations such as reordering or highlighting have limited effects. At the same time, many consumers close the consent window to avoid making an explicit choice, but they misunderstand what happens when they do so. Combined with the time costs of repeated consent decisions, these findings raise a central question: how should consent be designed to best serve consumer welfare?

Our welfare analysis shows meaningful gains from removing nudges and clarifying defaults. However, the largest benefits arise from replacing site-specific consent banners with browser-level consent management. We estimate that this policy increases consumer welfare by $3.67 per user-week relative to the optimal banner interface and by $4.27 relative to the U.S. status quo. While modest at the individual level, these gains aggregate to substantial societal benefits across internet users. Consistent with these findings, Global Privacy Control is gaining traction. It is supported by U.S. state privacy laws and recent EU reform proposals,[36] and is already implemented in browsers such as Firefox, Brave, DuckDuckGo, and Chrome extensions like Disconnect.

Our revealed-preference approach assumes that in the absence of nudges, observed choices reflect true preferences given users' beliefs. This assumption may be violated if users do not fully understand the consequences of cookie tracking. While we account for mistaken beliefs about banner closure in our welfare calculations, other information gaps

[36] See https://www.nelsonmullins.com/insights/alerts/privacy_and_data_security_alert/all/get-to-know-the-global-privacy-control-gpc-state-attorneys-general-remind-consers-of-their-right-to-opt-out-of-the-sale-or-sharing-of-personal-information and https://brusselssignal.eu/2025/11/eu-proposes-to-loosen-ai-and-data-privacy-rules-cut-cookie-banners (accessed November 2025).

may remain. Survey evidence suggests that participants have a basic understanding of cookies, but additional information could still affect their choices.

Although our experiment is more naturalistic than typical lab studies, it still differs from a fully natural environment in several aspects. First, self-selection into the study may limit external validity. Reassuringly, a follow-up survey comparing our participants with U.S. and EU samples finds no large differences in privacy valuations or preferences for consent designs. Second, participants' awareness of being studied and the use of standardized consent interfaces may affect behavior. If users are more attentive in our setting and standardized designs make choice frictions more salient, our estimates likely represent an upper bound on deliberation and a conservative estimate of the effects of defaults and choice frictions. Finally, our focus on short-term behavior leaves open the possibility of longer-term adaptation.

Our sample is also limited to Chrome users residing in the U.S., excluding individuals who rely on privacy-focused browsers such as Safari, Firefox, or DuckDuckGo, who may be more privacy-sensitive. While this selection may affect estimates of optimal default settings, they are unlikely to materially alter our main conclusions on the benefits of removing obstruction and moving to browser-level consent. Moreover, users of privacy-oriented browsers effectively make a browser-level privacy choice, consistent with our finding that centralized consent mechanisms can improve outcomes relative to site-by-site consent.

Finally, our analysis focuses on consumer welfare, and a full assessment of the supply-side and competitive effects of consent policies remains an important open question. In particular, while we find that GPC performs well from a consumer welfare perspective, its implications for competition are less clear. Understanding these competitive effects is an important direction for future research.

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Appendix A. Additional Tables and Figures

TABLE A.1. Websites Browsed during the Survey Phase

Domain	Domain Rank
facebook.com	3
youtube.com	8
amazon.com	28
yahoo.com	41
ebay.com	185
weather.com	325
duckduckgo.com	413
target.com	631
espn.com	278
etsy.com	301
nytimes.com	119
appleinsider.com	6,319
seattletimes.com	3,349
stockx.com	4,547
funnyordie.com	16,437
turo.com	16,272
semafor.com	28,266
thomannmusic.com	90,809
truewerk.com	348,372
merrysky.net	1,000,001

Notes: This table lists the websites that study participants are asked to visit at the beginning of the study. The second column shows domain rank, a proxy for site popularity. The websites are presented to participants in random order.

TABLE A.2. Number of Participants across the Experimental Funnel

Stage	N	Percent	Avg Age	Female (%)	Bachelor+ (%)	Income >$75k (%)
1) Start Survey	1227	100	37.73	46.4	53.4	42
2) Eligible for Study	917	74.7	37.73	46.4	54.6	43.4
3) Study Consent	877	71.5	37.73	46.4	54	43.4
4) Finished Survey	807	65.8	37.72	46.3	54.5	43.1
5) Have Cookie Choice Data	767	62.5	37.85	45.8	54.5	43
6) Choice for 15+ Domains	687	56	38.07	45.1	53.3	42.5
7) Choice for 1+ Organic Domain	602	49.1	38.41	45.5	52.2	43.9
8) Main Analysis Sample*	563	45.9	37.97	45.3	52	43.9
9) Finished Endline Survey	484	39.4	37.74	44.6	50.4	41.9

Notes: This table presents the number of study participants at every step of the study and main demographic characteristics (age, gender, share with at least a bachelor's degree, share with annual income above $75,000. $*$: The main analysis sample in the second-to-last line restricts attention to users who have treatment assignment to either the 10-Minute or 60-Minute group, and for whom we observe at least one cookie selection both during and after the survey.

TABLE A.3. Participants' Characteristics across Three Steps of the Funnel

Characteristic	Finished Survey	Main Sample	Finished Study
N	807	563	484
Panel A: Demographics			
Age (years)	37.72	37.97	37.74
Female (%)	46.3	45.3	44.6
Bachelor's or Above (%)	54.5	52.0	50.4
Income > $75,000 (%)	43.1	43.9	41.9
Panel B: Pre-Study Browsing			
Any Browsing History Data (%)	94.9	96.6	96.5
Browsing Visits Per Day (Prior Week)	338.5	385	395.4
Unique Domains Visited (Prior Week)	47.72	53.27	52.17
Active Browsing Days (Prior Week)	6.20	6.49	6.49
Panel C: Survey Familiarity			
Normally Visit (% of Survey Sites)		33.5	33.1
Heard Of (% of Survey Sites)		66.9	66.9
Panel D: Survey Cookie Choices			
Accept All (%)		60.9	61.1
Reject All (%)		23.5	23.7
Accept Some (%)		4.2	4.0
Close Window (%)		11.5	11.2
Panel E: Survey Responses			
Privacy Meaningful (%)			25.2
Believes Close Rejects Cookies (%)			60.7
Value Privacy (1-4)			3.28

Notes: This table summarizes participant characteristics across three samples: individuals who completed the initial survey, those included in the main analysis sample, and those who completed the full study (see lines 4, 8, and 9 of Table A.2). Each panel corresponds to a distinct subset of the available information on participants.

TABLE A.4. Choice Variation Decomposition (Outcome: "Accept All")

	Base	With Covariates
Intercept	0.625***	0.510***
	(0.014)	(0.072)
Acc-Rej-Set	-0.002	-0.005
	(0.008)	(0.008)
Acc-Set	0.036***	0.033***
	(0.008)	(0.008)
Rej-Acc-Set	-0.026**	-0.030***
	(0.008)	(0.008)
Rej-Set	-0.475***	-0.480***
	(0.008)	(0.008)
Set-Acc-Rej	-0.031***	-0.032***
	(0.008)	(0.008)
SD (Participant)	0.304	0.304
SD (Domain)	0.058	0.046
SD (Residual)	0.343	0.342
Num.Obs.	23685	23190

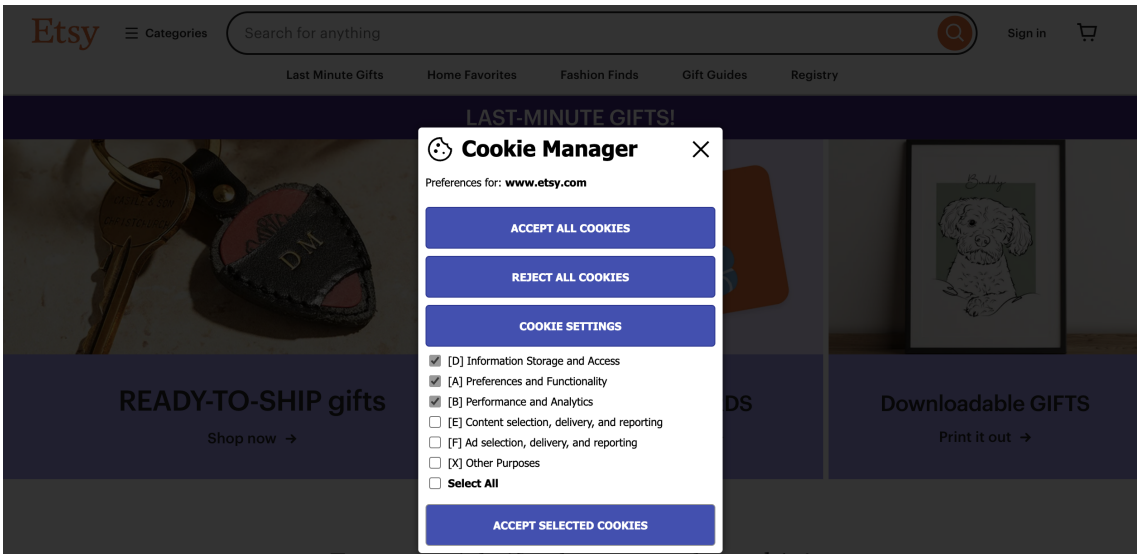
Notes: This table presents estimates of treatment effects models where random effects for participant and domain are included in the regression. The second column adds controls for website characteristics, demographics, and privacy beliefs presented in Table 3. The outcome is a dummy for whether a user accepts all cookies.

TABLE A.5. Consumer Surplus Under Counterfactual Policies Using Median Time ($, Weekly)

Counterfactual	Average	Type 1 Acceptors	Type 2 Rejectors	Type 3 Discerners
U.S. Status Quo	1.96	2.29	-2.05	3.47
EU Norm	2.3	2.22	1.1	2.97
Optimal Banner Interface	2.46	2.7	-0.51	3.58
Optimal Banner Interface (with correct beliefs)	2.82	2.99	0.51	3.7
Global Accept	2.66	3.64	-1.91	3.64
Global Reject	1.98	1.83	1.96	2.17
Global Privacy Control	3.34	3.64	1.96	3.64

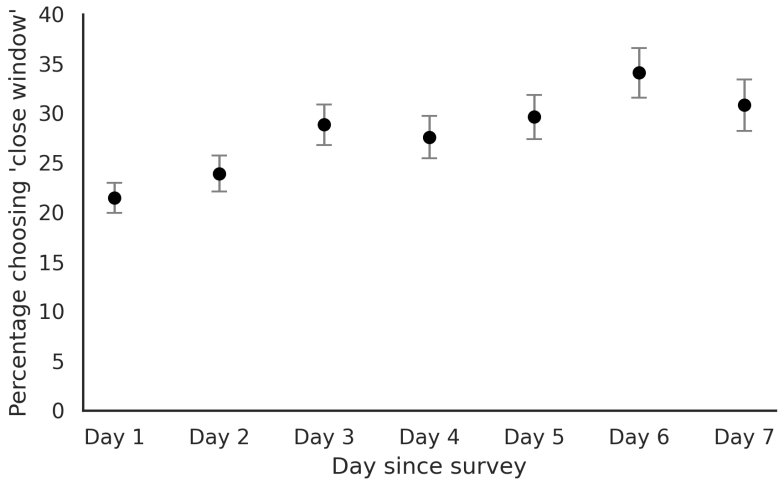
Notes: The table reports consumer surplus per week per user in dollar units using median time spent rather than average time spent. All policies are defined analogous to Table 8.

FIGURE A.1. Cookie Manager's User Interface



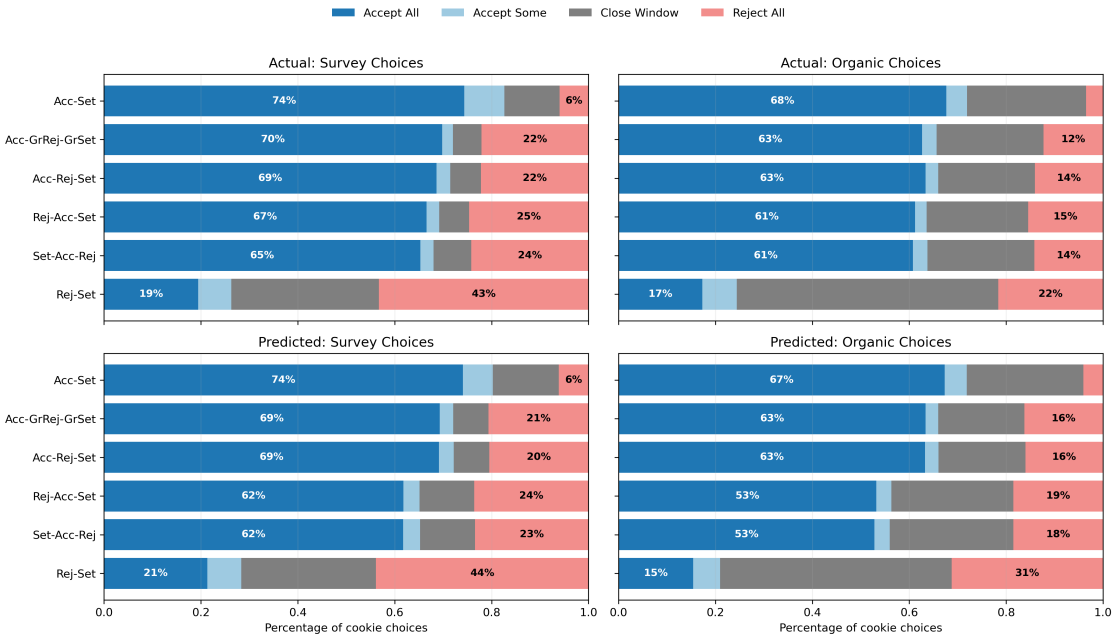
Notes: Consent interface when a user clicks "cookie settings". The available blue buttons vary by treatment: for example, in the "accept-settings" design, the blue "Reject All Cookies" button does not show up when a user clicks "settings." However, the list of specific cookies within the settings menu is identical across treatments, meaning they can always choose to reject or select all cookies upon coming to the "settings" menu.

FIGURE A.2. Frequency of Closing Windows Over the Study Period



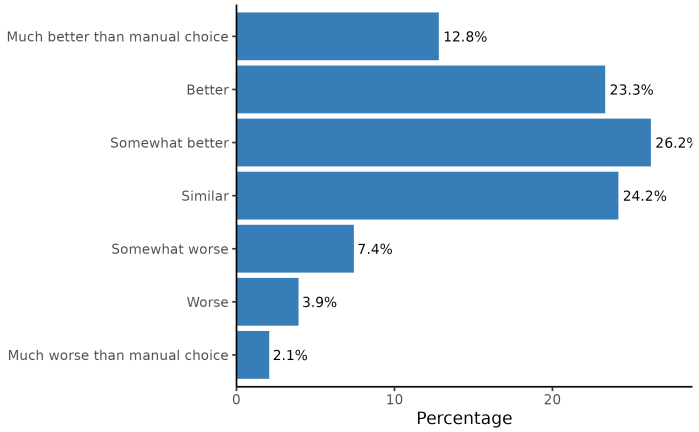
Notes: This figure shows the percentage of times participants close the consent window without choosing on each day during the organic phase. Error bars represent 95% confidence intervals.

FIGURE A.3. Actual versus Predicted Choice Shares



Notes: This figure displays the proportions of cookie choices across banner design treatments. The top panel presents actual choices (identical to Figure 2). The bottom panel presents predicted choices from our structural estimates in Table 7.

FIGURE A.4. User Preferences for Global Privacy Control



Notes: This figure shows how respondents in the endline survey value a global privacy control tool that automatically applies their preferred cookie choices across websites, relative to answering cookie banners manually on each site.

Online Appendix to "Designing Consent: Choice Architecture and Consumer Welfare in Data Sharing"

Appendix B. Covariate Balance

TABLE B.1. Covariate Balance Check for Randomization of Consent Banner Designs

	Age (1)	Female (2)	Bachelor's or Above (3)	Domain Rank (Log 10) (4)
Constant	38.720***	0.438***	0.182***	3.579***
	(0.196)	(0.007)	(0.006)	(0.023)
Acc-GrRej-GrSet	0.061	-0.004	-0.009	-0.034
	(0.281)	(0.011)	(0.008)	(0.033)
Acc-Rej-Set	-0.040	0.013	0.005	-0.004
	(0.281)	(0.011)	(0.008)	(0.033)
Acc-Set	-0.023	-0.001	-0.007	-0.023
	(0.280)	(0.010)	(0.008)	(0.033)
Rej-Acc-Set	0.343	0.011	-0.008	0.013
	(0.284)	(0.011)	(0.008)	(0.033)
Rej-Set	0.312	-0.008	-0.011	-0.045
	(0.286)	(0.011)	(0.008)	(0.033)
R^2	0.000	0.000	0.000	0.000
Observations	26,278	26,278	26,773	26,773

Notes: Banner design is randomized at the user X site level. *$p < 0.05$, ** $p < 0.01$; *** $p < 0.001$.

TABLE B.2. Covariate Balance Check for Banner Frequency Randomization

	# Survey Domains (1)	Age (2)	Female (3)	Bachelor's or Above (4)
Constant	18.537***	37.460***	0.443***	0.563***
	(0.222)	(0.703)	(0.027)	(0.027)
10 Min Pop-up	0.123	1.560	0.039	-0.076·
	(0.307)	(1.037)	(0.040)	(0.039)
R^2	0.000	0.004	0.001	0.006
Observations	656	638	638	656

Notes: Banner frequency is randomized at the user level. We therefore exclude domain rank during the organic browsing, but include the number of banners exposed at the survey stage for covariate balance checks. *$p < 0.05$, ** $p < 0.01$; *** $p < 0.001$.

Appendix C. Additional Reduced-Form Results

TABLE C.1. Types of Cookies Accepted among People Making Granular Choices

Cookie Type	Percentage Selected
Preferences and functionality	0.826
Information storage and access	0.627
Performance and analytics	0.601
Content selection, delivery, and reporting	0.390
Ad selection, delivery, and reporting	0.070
Other purposes	0.048

Notes: Percentage of different types of cookies selected among those who selectively accept some cookies but not all.

TABLE C.2. Differences Between Participants Who Always Accept, Always Reject, and Make Discerning Choices in the Baseline Condition

Variable	Mean: Never Accept	Mean: Some Accept	Mean: Always Accept	F-value	p-value
Age	38.81	38.22	37.77	0.19	0.83
Female	0.48	0.41	0.45	0.61	0.55
Bachelor's or Above	0.63	0.5	0.51	2.08	0.13
Income > $75,000	0.43	0.48	0.42	0.79	0.45
Prior Domains Visited	52.27	57.5	50.87	1.43	0.24

Notes: "Never Accept" and "Always Accept" indicate the participants who never and always choose "accept all" while in the baseline condition, excluding instances of closing window. In this table, all choice probabilities are calculated within the baseline design treatment, thus choice variations in the "Some Accept" group reflect variation across sites, excluding the influence of choice architecture on choice.

TABLE C.3. Cookie Choice by Experimental Condition (Survey vs. Organic)

	Survey				Organic			
	Accept All	Reject All	Close Window	Accept Some	Accept All	Reject All	Close Window	Accept Some
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Acc-Set	0.083***	-0.178***	0.039***	0.056***	0.054***	-0.094***	0.019	0.021**
	(0.013)	(0.015)	(0.009)	(0.011)	(0.012)	(0.013)	(0.013)	(0.007)
Acc-GrRej-GrSet	0.035***	-0.017	-0.012	-0.007	0.031**	-0.016	-0.020*	0.005
	(0.010)	(0.010)	(0.006)	(0.004)	(0.011)	(0.008)	(0.010)	(0.003)
Acc-Rej-Set	0.020	-0.007	-0.010	-0.003	0.038**	-0.001	-0.041***	0.005
	(0.011)	(0.009)	(0.006)	(0.005)	(0.011)	(0.007)	(0.010)	(0.003)
Rej-Acc-Set	0.003	0.014	-0.012*	-0.005	0.004	0.020*	-0.023*	-0.002
	(0.010)	(0.010)	(0.006)	(0.005)	(0.011)	(0.009)	(0.010)	(0.004)
Rej-Set	-0.464***	0.193***	0.233***	0.038***	-0.427***	0.086***	0.302***	0.039***
	(0.020)	(0.017)	(0.018)	(0.008)	(0.024)	(0.012)	(0.023)	(0.007)
Benchmark group mean	0.65	0.24	0.08	0.03	0.61	0.14	0.22	0.03
R^2	0.646	0.579	0.562	0.413	0.571	0.522	0.494	0.499
Observations	11,075	11,075	11,075	11,075	12,610	12,610	12,610	12,610
Participant fixed effects	✓	✓	✓	✓	✓	✓	✓	✓
Domain Cat. fixed effects	✓	✓	✓	✓	✓	✓	✓	✓

Notes: Regression results of Equation 1 for three outcomes: accept all cookies, reject all cookies, and close the window without making a choice. This table splits results from Table 2 into survey and organic choices: survey choices are presented in columns 1 through 4, organic choices are in columns 5 through 8. For the survey estimates, we pre-registered controlling for site fixed effects instead of site category fixed effects. Those estimates are available in Appendix C.4 and are undistinguishable from those in the first four columns of this table. Standard errors clustered at the participant level. *$p < 0.05$, ** $p < 0.01$; *** $p < 0.001$.

TABLE C.4. Cookie Choice by Experimental Condition (Website Fixed Effects)

	Accept All	Reject All	Close Window	Accept Some
	(1)	(2)	(3)	(4)
Acc-Set	0.085***	-0.179***	0.039***	0.056***
	(0.013)	(0.015)	(0.009)	(0.011)
Acc-GrRej-GrSet	0.037***	-0.018	-0.012	-0.007
	(0.010)	(0.009)	(0.006)	(0.004)
Acc-Rej-Set	0.021*	-0.008	-0.010	-0.003
	(0.011)	(0.008)	(0.006)	(0.005)
Rej-Acc-Set	0.004	0.012	-0.012*	-0.004
	(0.010)	(0.010)	(0.006)	(0.005)
Rej-Set	-0.464***	0.192***	0.234***	0.038***
	(0.020)	(0.017)	(0.018)	(0.008)
Benchmark group mean	0.65	0.24	0.08	0.03
R^2	0.651	0.584	0.563	0.414
Observations	11,075	11,075	11,075	11,075
Participant fixed effects	✓	✓	✓	✓
Domain fixed effects	✓	✓	✓	✓

Notes: This table presents regression estimates similar to the first four columns of Table C.3. The only difference is that here we control for website fixed effects rather than only website category fixed effects. Standard errors clustered at the participant level. *$p < 0.05$, ** $p < 0.01$; *** $p < 0.001$.

TABLE C.5. Effects of Choice Architecture by Website Characteristics

a. Effects of Choice Architecture by Prior Visits

	Accept All (1)	Reject All (2)	Close Window (3)
Has Prior Visit × Rej-Acc-Set	0.028	-0.009	-0.002
	(0.020)	(0.015)	(0.015)
Has Prior Visit × Acc-Set	0.007	0.038**	-0.030
	(0.018)	(0.014)	(0.016)
Has Prior Visit × Acc-GrRej-GrSet	0.011	-0.005	0.001
	(0.018)	(0.013)	(0.015)
Has Prior Visit × Acc-Rej-Set	0.017	-0.008	-0.008
	(0.019)	(0.014)	(0.016)
Has Prior Visit × Rej-Set	0.011	-0.044*	0.022
	(0.022)	(0.018)	(0.020)
Has Prior Visit	0.015	-0.029*	0.006
	(0.015)	(0.012)	(0.012)
R^2	0.517	0.471	0.421
Observations	23,685	23,685	23,685
Condition fixed effects	✓	✓	✓
Participant fixed effects	✓	✓	✓
Domain Cat. fixed effects	✓	✓	✓

b. Effects of Choice Architecture by Domain Popularity

	Accept All (1)	Reject All (2)	Close Window (3)
Domain Rank (Log 10) × Rej-Acc-Set	-0.002	-0.001	0.001
	(0.005)	(0.004)	(0.004)
Domain Rank (Log 10) × Acc-Set	-0.007	0.004	0.005
	(0.005)	(0.004)	(0.004)
Domain Rank (Log 10) × Acc-GrRej-GrSet	0.000	-0.003	0.002
	(0.005)	(0.004)	(0.004)
Domain Rank (Log 10) × Acc-Rej-Set	-0.002	0.002	0.000
	(0.005)	(0.004)	(0.004)
Domain Rank (Log 10) × Rej-Set	0.005	-0.015**	0.013*
	(0.006)	(0.005)	(0.006)
Domain Rank (Log 10)	-0.010*	0.007*	0.003
	(0.004)	(0.003)	(0.003)
R^2	0.518	0.470	0.422
Observations	23,685	23,685	23,685
Condition fixed effects	✓	✓	✓
Participant fixed effects	✓	✓	✓
Domain Cat. fixed effects	✓	✓	✓

Notes: The table shows regression estimates similar to Table 3, Panel *b*, except that the banner design treatment dummies are also interacted with whether the site is visited before (Panel a) and site rank (Panel b). "Condition fixed effect" refers to indicator variables for the 6 banner design conditions. *$p < 0.05$, ** $p < 0.01$; *** $p < 0.001$.

Appendix D. Decomposing Reduced-Form Effects by Type of Choice Architecture

To quantify the effect of specific choice architecture on privacy choices, we identify the presence of reordering, highlighting, and obstructions in our treatment banners. This means estimating the following regression:

$$y_{ij} = \gamma_{reject\ hidden_{ij}} + \gamma_{accept\ hidden_{ij}} + \gamma_{accept\ on\ top_{ij}} + \gamma_{reject\ on\ top_{ij}} + \gamma_{highlight\ accept_{ij}} + \mu_i + \nu_{c(j)} + \epsilon_{ij}, \tag{D.1}$$

where i indexes study participants and j indexes websites. We include participant fixed effects μ_i and website category fixed effects $\nu_{c(j)}$. The γ indicators capture the specific nudges present in the consent interface for user i on website j. We focus on three outcomes: accepting all cookies, rejecting all cookies, and closing the window without making an active choice.

Table D.1 presents the results. Deliberate obstruction emerges as the most effective design. Hiding the "reject all" button from the main screen reduces rejection rates by 13 percentage points (a 67% decrease). Hiding the "accept all" button has a similar pronounced effect, reducing acceptance rates by 45 percentage points (a 71% decline). In contrast, visual manipulations—i.e., reordering options and highlighting "accept all"—have limited effects on user choices, with small and statistically insignificant coefficients.

TABLE D.1. Cookie Selections by Choice Architecture

	Accept All (1)	Reject All (2)	Close Window (3)
Reject Hidden	0.037***	-0.127***	0.055***
	(0.009)	(0.011)	(0.008)
Accept Hidden	-0.448***	0.119***	0.289***
	(0.017)	(0.011)	(0.017)
Accept Top	0.029***	-0.002	-0.029***
	(0.008)	(0.006)	(0.006)
Reject Top	0.005	0.020**	-0.022**
	(0.008)	(0.007)	(0.007)
Highlight Accept	0.001	-0.013	0.014*
	(0.008)	(0.007)	(0.006)
Benchmark group mean	0.63	0.19	0.16
R^2	0.517	0.469	0.421
Observations	23,685	23,685	23,685
Participant fixed effects	✓	✓	✓
Domain Cat. fixed effects	✓	✓	✓

Notes: Regression results of Equation D.1 for three outcomes: accept all cookies, reject all cookies, and close window without making a choice. *$p < 0.05$, ** $p < 0.01$; *** $p < 0.001$.

Appendix E. Choice Fatigue

In this section, we examine whether the attention users pay to choices changes as they see more consent banners. We compare the differences in choices between our 10-minute and 60-minute treatments to show that there is no choice fatigue when we increase the frequency of banners.

The 10-minute treatment sees our banners in 53% of the domains they visited, while the 60-minute treatment sees these banners in 30% of the domains. Given this difference, we can examine whether the frequency of choice types differs between the two conditions. We estimate the effects of this treatment in the following regression specification.

$$y_{ij} = \beta_{10\ minutes} + \gamma * time\ in\ study_{ij} + \nu_{c(j)} + \epsilon_{ij}. \tag{E.1}$$

The baseline condition is when a user sees the banner every 60 minutes, while the alternative condition displays a banner every 10 minutes. We also include the time a user has been in the study after the survey phase.[37]

Table E.1 displays the results. We highlight two findings. First, we do not find a differential impact of banner frequency on data-sharing choices, whether it is the acceptance rate or the inclination to close banners. These null effects are precisely estimated, as the 95% confidence interval excludes effects greater than 7%. However, we acknowledge the caveat that the difference between exposing to banners 30% vs. 53% of the time may not be large enough compared to, say, comparing banner exposure between 30% and 100% of the time.

Second, time spent in the study is associated with disengagement in privacy choices. For each additional day in the study, participants increase the inclination to close the banner by two percentage points. We note that time in the study is not randomly allocated and could be correlated with underlying consumer characteristics and privacy preferences. To address this concern, in Table E.2 we add individual and hour-of-the-day fixed effects, and control for the order of the domain visit. With these additional controls, domains visited at a later time (*Vitit Order/10* correlate with acceptance and increases in close-out, suggesting that participants reduce their engagement with the study over repeated banner interactions.

Lastly, the inclination to close windows stabilizes over the 7-day period (see Figure A.2). This pattern provides assurance that our study, especially towards the end, captures the steady-state behavior of participants as they get accustomed to the banner exposure.

[37] Adding this covariate does not affect whether we detect any treatment effects.

TABLE E.1. Fatigue in Cookie Choices During Organic Browsing

	Accept All (1)	Reject All (2)	Close Window (3)
10 Min Pop-up	0.009	-0.005	-0.003
	(0.037)	(0.024)	(0.031)
Time in Study (Days)	-0.009*	-0.008**	0.017***
	(0.004)	(0.003)	(0.004)
Domain Rank (Log 10)	0.010**	-0.008**	-0.002
	(0.004)	(0.003)	(0.003)
Pre-Exp Visit	0.041**	-0.031*	-0.015
	(0.015)	(0.013)	(0.013)
R^2	0.008	0.007	0.009
Observations	12,610	12,610	12,610
Domain Cat. fixed effects	✓	✓	✓

Notes: This table shows estimates of Equation E.1, where '10 Min Pop-up' is an indicator for whether the user was in the treatment where banners occurred at a frequency of once every 10 minutes. *$p < 0.05$, ** $p < 0.01$; *** $p < 0.001$.

TABLE E.2. Fatigue in Cookie Choices During Organic Browsing (Additional Fixed Effects)

	Accept All (1)	Reject All (2)	Close Window (3)
Visit Order / 10	-0.020*	-0.004	0.023*
	(0.009)	(0.004)	(0.009)
Time in Study (Days)	0.000	-0.003	0.005
	(0.005)	(0.003)	(0.005)
R^2	0.469	0.501	0.439
Observations	12,610	12,610	12,610
Domain Cat. fixed effects	✓	✓	✓
Participant fixed effects	✓	✓	✓
Hour fixed effects	✓	✓	✓

Notes: This table estimates a variant of Equation E.1, which removes the banner frequency treatment and adds the order of which a domain is visited ("Visit Order") and additional fixed effects. *$p < 0.05$, ** $p < 0.01$; *** $p < 0.001$.

Appendix F. Alternative Structural Model Estimates

This appendix presents alternative model specifications, with 2-type and 4-type latent classes, respectively in Table F.1 and Table F.2.

TABLE F.1. Parameter Estimates Across Consumer Subsets (2-Class Estimate)

Parameter	Explanation	Estimates	
Panel A: Utility Parameters			
Type-Specific Parameters		Type1	Type2
θ_{acc}	Utility of accepting all	2.100***	-1.740
		(0.169)	(1.040)
θ_{set}	Utility of granular choice	0.570***	0.280***
		(0.125)	(0.074)
C_{set}	Cost of clicking settings	2.260***	2.250***
		(0.111)	(0.174)
δ_{top}	Effect of ranking on top	0.283***	-0.388
		(0.066)	(0.577)
δ_{gray}	Effect of being grayed out	4.260	-11.000*
		(2.810)	(5.480)
Pooled Parameters			
κ_1	Utility of closing window	0.725***	
		(0.043)	
κ_2	Inclination of closing over time	-1.730***	
		(0.056)	
ρ_{unc}	Pr(accept | close) for uncertain	0.328***	
		(0.078)	
ρ_{miss}	Pr(accept | close) for missing	0.705***	
		(0.123)	
Panel B: Latent Class Probability Parameters			
λ^1	Intercept	1.140***	
		(0.272)	
λ^2	Bachelor degree or higher	-0.387***	
		(0.056)	
λ^3	Nr. domains in browsing history	0.087	
		(0.055)	
Share of Users		0.741	0.259
Log likelihood		-23,203.95	
Bayesian information criterion		46,579.13	

Notes: The table presents parameter estimates from the utility model assuming a 2-type latent class model. Otherwise, the table is identical to Table 7. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

TABLE F.2. Parameter Estimates Across Consumer Subsets (4-Class Estimate)

Parameter	Explanation	Estimates			
Panel A: Utility Parameters					
Type-Specific Parameters		Type1	Type2	Type3	Type4
θ_{acc}	Utility of accepting all	2.650***	-5.350	-5.010*	1.190***
		(0.181)	(21.400)	(2.290)	(0.081)
θ_{set}	Utility of granular choice	0.430	-0.691	-0.288	0.590*
		(0.356)	(0.784)	(1.200)	(0.287)
C_{set}	Cost of clicking settings	2.900***	2.490	2.650	1.660***
		(0.201)	(3.530)	(1.540)	(0.189)
δ_{top}	Effect of ranking on top	0.434***	0.147	-1.250	0.086
		(0.104)	(0.275)	(30.600)	(0.079)
δ_{gray}	Effect of being grayed out	-1.520	-2.010	-1.660	-0.118
		(26.600)	(10.700)	(26.600)	(1.270)
Pooled Parameters					
κ_1	Utility of closing window	0.748***			
		(0.048)			
κ_2	Inclination of closing over time	-1.660***			
		(0.070)			
ρ_{unc}	Pr(accept | close) for uncertain	0.350***			
		(0.048)			
ρ_{miss}	Pr(accept | close) for missing	0.458***			
		(0.137)			
Panel B: Latent Class Probability Parameters					
λ^1	Intercept	1.130***	-3.490	0.582**	
		(0.131)	(3.220)	(0.188)	
λ^2	Bachelor degree or higher	-0.821***	0.365	0.886*	
		(0.225)	(6.310)	(0.359)	
λ^3	Nr. domains in browsing history	0.049	-0.095	1.520***	
		(0.047)	(0.793)	(0.340)	
Share of Users		0.453	0.006	0.374	0.167
Log likelihood		-23,020.13			
Bayesian information criterion		46,372.65			

Notes: The table presents parameter estimates from the utility model assuming a 4-type latent class model. Otherwise, the table is identical to Table 7. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix G. Cross-Country Comparison from Stated-Preference Survey

In fall 2025, we ran a supplemental survey across three Prolific populations: participants from our main study, U.S. respondents who did not participate our main study, and EU respondents. The survey consisted of three blocks. The first block collected demographic information.[38] The second block focused on privacy attitudes: what privacy means to respondents and their perceptions of what advertisers know about them. The third block involved three pairwise comparisons of consent interfaces: (1) acc–set versus acc–rej–set, (2) acc–rej–set versus acc–rej–set with all settings options immediately visible, and (3) respondents' preferred banner versus global privacy control (a similar comparison is contained in the outtake survey of the main study). We recruited 320 participants from our main study, 471 additional participants residing in the U.S., and another 471 participants from the EU. The 320 participants from our prior study are statistically indistinguishable from main study participants who did not answer the new survey (see Table G.1).

TABLE G.1. Selection into Follow-Up Survey

	Accept all (1)	Privacy importance (2)	Global Privacy Controls (3)	WTP for Global Privacy Controls (4)
Constant	0.602***	3.265***	0.872***	3.814***
	(0.017)	(0.041)	(0.091)	(0.465)
Respondent to New Survey	-0.038	0.034	0.055	0.404
	(0.027)	(0.059)	(0.127)	(0.706)
Mean of Y	0.58	3.28	0.9	4.01
R^2	0.004	0.001	0.000	0.001
Observations	563	483	484	484

Notes: The table reports linear regressions comparing main study participants who completed the Fall 2025 follow-up survey with those who did not. All outcomes are measured at the time of the main study. The key independent variable is an indicator for completing the follow-up survey. Column (1) corresponds to the average share of accept all clicks recorded by the browser extension during the main study. Column (2) corresponds to responses on a five-point Likert scale to the question: "Privacy means different things to different people. In thinking about all of your online browsing, please state how important it is for you to be in control of who can get information about you." A value of 5 corresponds to *very important*. Column (3) is a dummy for whether the respondent said they prefer global privacy control to website-level consent banners. Column (3) corresponds to the willingness to pay for a global-privacy-control tool. The number of observations in column (1) reflects the total number of participants in the main study (line 9 in Table A.2). All other columns include only participants who completed the outtake survey of the main study (line 10 in Table A.2). * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

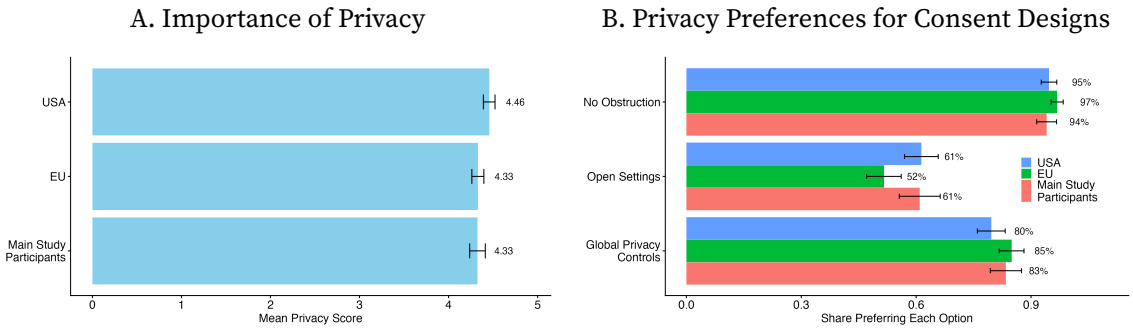
To test for order and framing effects, participants were randomly assigned to one of four conditions in a factorial design that varied (a) the order of privacy attitudes and consent-comparison blocks, and (b) whether valuations were elicited as willingness-to-pay (WTA) or willingness-to-accept (WTP). We don't find meaningful effects of question order

[38] Requesting demographic information allows us to account for how demographic compositions affect the differences in the responses across samples. This is necessary because by requiring English fluency to complete the survey, our EU sample skews younger and more highly educated.

on their stated valuations. In contrast, framing matters substantially, generating a wide gap between reported WTAs and WTPs.

There are only limited differences between prior study participants and the broader survey sample. The left panel of Figure G.1 reports respondents' stated valuation for online privacy. On average, participants assign a value of 4.38 (the maximum value of 5 denotes *very important*), with only modest variation across consumer groups.

FIGURE G.1. Preferences by Consumer Groups



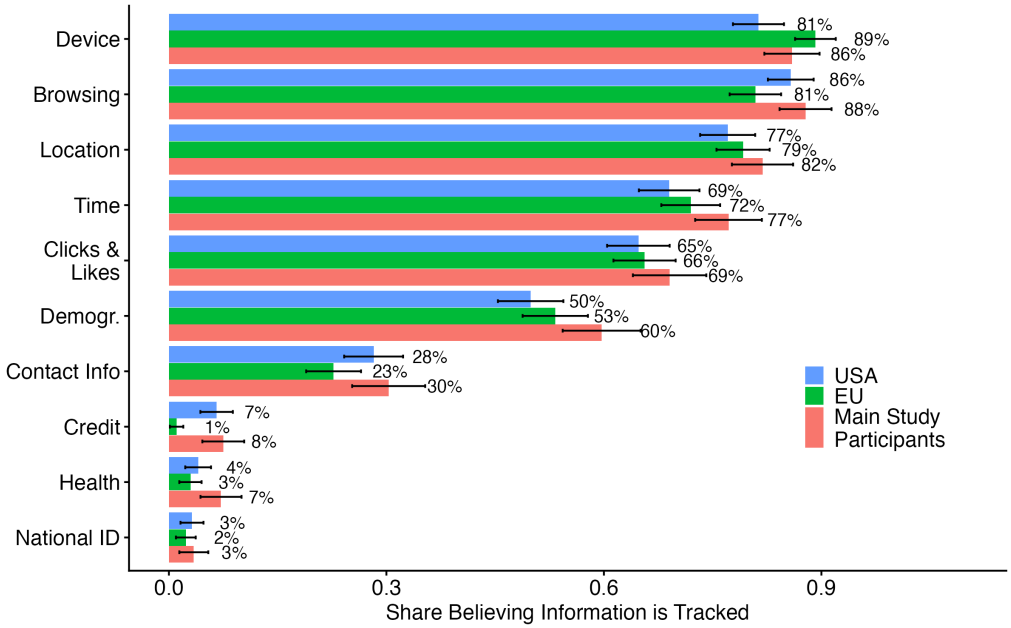
Notes: Figure plots privacy preferences across three consumer groups: participants in our main study, U.S. survey respondents, and EU survey respondents. All respondents were surveyed in Fall 2025. The left panel reports the average response to a five-point Likert-scale question: "Privacy means different things to different people. In thinking about your online browsing, how important is it for you to be in control of who can access information about you?" A value of 5 corresponds to very important. The right panel reports the share of respondents who prefer (1) a banner with no obstruction over one that hides the "reject all" option, (2) a banner that displays all settings upfront over one that requires an additional click, and (3) global privacy control over their preferred website-level consent banner. Bars indicate 95% confidence intervals.

There is also a striking similarity across groups in the types of information respondents believe are tracked about them online (see Figure G.2). Browsing history and device information are the most commonly mentioned, followed by location data and time spent on apps or websites, and then clicks and likes on social media. Demographics, contact information, credit data, health information, and national ID details are mentioned less for all groups.

To assess whether preferences for consent designs differ meaningfully across respondents, we compare their stated choices over alternative banners. Consumers express a clear dislike for designs that deliberately obstruct choices: 95.3% prefer the accept–reject–settings banner over the accept–settings version. Preferences regarding the visibility of cookie settings are mixed, with 57.6% favoring banners that display all settings on the first page. Support for global privacy control is higher, with 82.6% of respondents rating them comparable (9.3%) or better (73.3%) than their preferred site-level consent banner.

The right panel of Figure G.1 shows that these patterns are qualitatively similar across groups. If anything, European respondents are more likely to prefer designs that make choices easier (e.g., less likely to prefer exposing all settings immediately and more likely to prefer GPC) relative to U.S. participants and our main-study sample.

FIGURE G.2. Belief By Consumer Groups



Notes: The figure reports the share of respondents who believe advertisers typically know various pieces of information about them. Each column corresponds to one option from the survey question, "As you browse the internet, which of the following do you think advertisers typically know about you?" with outcomes coded as binary indicators. Columns span (1) device type, (2) browsing history, (3) location, (4) time spent on websites/apps, (5) social-media interactions, (6) demographic information, (7) contact information, (8) credit score, (9) health information, and (10) national ID number; columns are ordered by frequency of selection. "USA" identifies non-main-study US respondents, "EU" identifies European respondents, "Main Study Participants" indicates respondents who took part in the main study. Error bars show 95% confidence intervals.

Appendix H. Survey Questions

This appendix presents the Qualtrics surveys used in the study:

- Intake.
- Outtake.
- Stated preference survey across three Prolific populations: main study participants, US consumers, and EU consumers.

Device Transfer

The rest of the survey needs to be done on a Chrome browser. Please copy the link of the survey and reopen it in a Chrome browser to continue.

First Page

Would you like to help us understand online behavior and privacy choices? We are a team of Harvard and Boston University researchers who study the internet and how it affects society.

The study will take 30 minutes of your time over the course of the next day. We will ask you to fill out two surveys, clear the cookie data stored in your browser, install a browser extension vetted by Harvard and Boston University and keep it installed for seven days, and share information about your online behavior. Click below if you want to know more and discover if you qualify!

Eligibility Questions

Do you live in the United States?

Yes
No

Are you over 18 years old?

Yes
No

What is 12 minus 4? Regardless of the correct answer, you should always select the option with the value "seven". This is an attention check question.

8
5
6
7

What is the language you primarily speak?

Spanish
English
Other (please specify)

Which browser do you primarily use?

Firefox
Others
Chrome
Internet Explorer
Safari
Microsoft Edge

What was your total household income before taxes during the past 12 months?

Less than $25,000
$25,000-$49,999
$50,000-$74,999
$75,000-$99,999
$100,000-$149,999
$150,000 or more
Prefer not to say

What is the highest level of education you have completed?

Some high school or less
High school diploma or GED
Some college, but no degree
Associates or technical degree
Bachelor's degree
Graduate or professional degree (MA, MS, MBA, PhD, JD, MD, DDS etc.)
Prefer not to say

Not Eligible

Thank you for your answers! Unfortunately, you do not qualify to participate in our study.
Can you please return your submission on Prolific?

Consent

Congratulations! You are qualified to participate in our study.

Study Overview

The following is a summary with key information to help you decide whether you want to participate.

Why am I being invited to take part in a research study?
We invite you to take part in this research study because you are an English-speaking resident of the United States who uses Chrome to browse the web.

What should I know about a research study?
Research studies are conducted to better understand the choices we make. Whether or not you take part is completely up to you. Your decision will not be held against you. You can ask all the questions you want before you decide. You can even agree to take part and later change your mind.

Why is this research being done?
We want to better understand the online experience of people like you, how companies obtain user consent for the collection and use of their data, and how this affects user browsing experience. We hope that the results of this research will help inform data privacy policy.

How long will the research last and what will I need to do?
The study will last several days, but we will only ask you for 30 minutes of your time. Everything we ask you to do to participate in this research can be done from the comfort of your home. If you choose to participate, we'll ask you to:

- Complete two surveys:
  - The first survey will ask you some questions about yourself and your online browsing behavior. It will also ask you to visit some websites and make privacy choices.
  - The second survey will ask you about your experience during the study.
- Install the Cookie Manager browser extension, which is an application we developed for this study. We'll have instructions for you. The Cookie Manager extension will record your behavior and may tweak the interfaces through which you make cookie selections.
- Keep the extension installed for seven days, until the extension prompts you to uninstall it.

Will I be compensated for participating in this research?
Yes. You will be paid $7.50 after completing the two surveys and keeping the Cookie Manager extension installed for several days.

Is there any way being in this study could be bad for me?
Since we may collect personal information, there is a risk of breach of confidentiality. We have worked hard to minimize this risk. For example, we will encrypt any data before storing it. Before accessing the data for analysis, we will also permanently delete all personal information that we may intentionally or unintentionally collect.

Will being in this study help me in any way?
We cannot promise any benefits to you or others from your taking part in this research. It is possible, however, that our tweaks to your online browsing lead to a better (or worse) online experience.

Detailed Information

Withdrawing from the Study.
You can leave the research at any time; your decision will not be held against you. We may use the data you have shared with us prior to withdrawing as part of the study. We will provide simple instructions for how you can withdraw. Researchers can remove you from the research study without your approval. Possible reasons for removal include not complying with instructions to install the browser extension or intentionally avoiding data tracking through the extension.

Privacy.
Data security and privacy are important to us. During the course of the study we may collect personal information. The personal information that we know we are collecting will be deleted immediately. Other personal information that we inadvertently collect will be stored but removed after we finish collecting data.

We cannot promise complete secrecy, although efforts will be made to limit the use and disclosure of your personal information. Data will be encrypted and stored on secure servers and cannot be accessed by anyone outside the research team. At no time will study information be available over any public or private network in an unencrypted state.

In the future, when we publish our research, we will post anonymized data from this study in a data repository so that other researchers can reproduce our results. By then, no information that can identify you personally will be available, to us or others. We will not sell data from the study or share data for any commercial or marketing purposes.

Who can I talk to?
If you have questions, concerns, or complaints, or think the research has hurt you, do not hesitate to reach the research team on Prolific or cookie.manager.study@gmail.com.

Please indicate below whether you agree to participate in the study. Agreeing to participate means you are willing to install Cookie Manager (our web browser extension) for seven days, and complete the two surveys.

I agree to participate
I do not agree to participate

Not consent

Thank you for letting us know you do not want to participate. **Can you please return your submission on Prolific?**

Email

Thank you for your willingness to participate in our study!

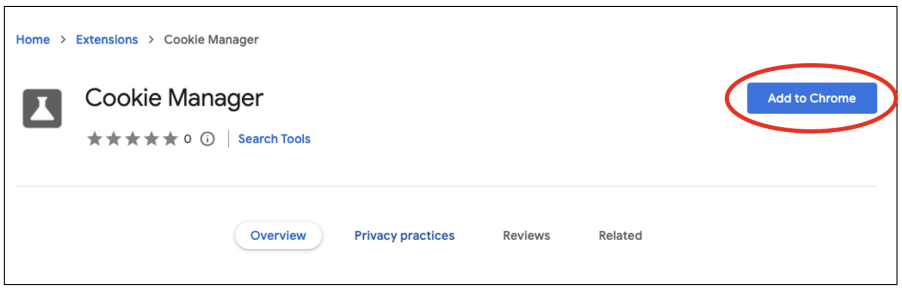
Next, we will ask you to install *Cookie Manager*, a browser extension we developed to identify website tracking and to enable simplified privacy consent dialogs.

After installing the extension, you will see a consent-request popup window whenever you visit a website for the first time. If you make a choice, the extension will try to pass on your choices to the website. In most cases, if the website has already been collecting consent from users, it will recognize your choice and decide whether to continue tracking you based on your choice.

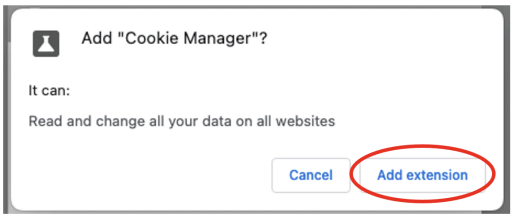
App Installation

Cookie Manager Installation Instructions.
To install Cookie Manager, please **use Chrome** on the computer that you are using for online shopping:

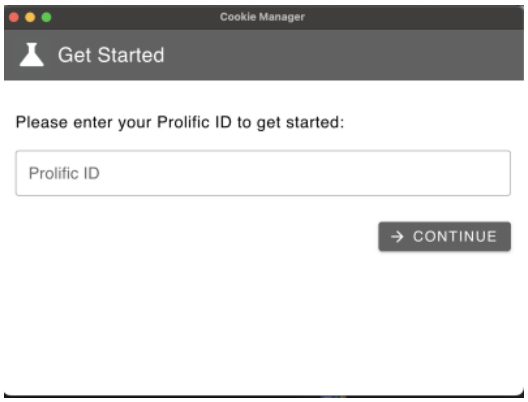
- Click here.
- Click “Add to Chrome.”



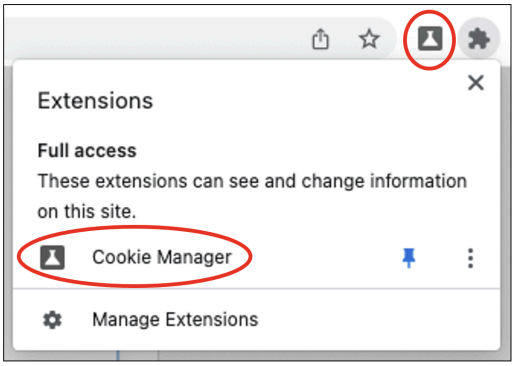
- When prompted, click “Add Extension.”



- You will be prompted to add your prolific id.



- You should now see the Cookie Manager icon on the top right corner of your browser. If you don’t see it, it may be hidden under the puzzle icon, which is visible in the upper right corner of the screenshot below.



- You are all set.

If you have trouble installing Cookie Manager, email us at cookie.manager.study@gmail.com and we will help you with additional instructions.

Were you able to successfully install the extension?

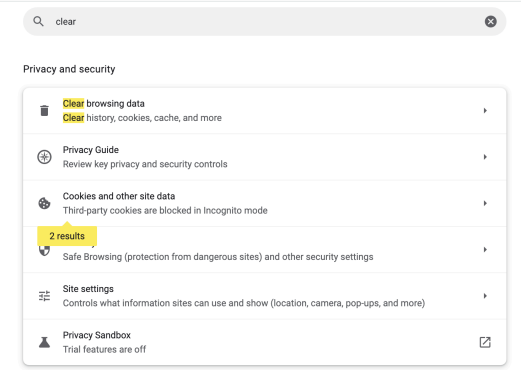
Yes

No

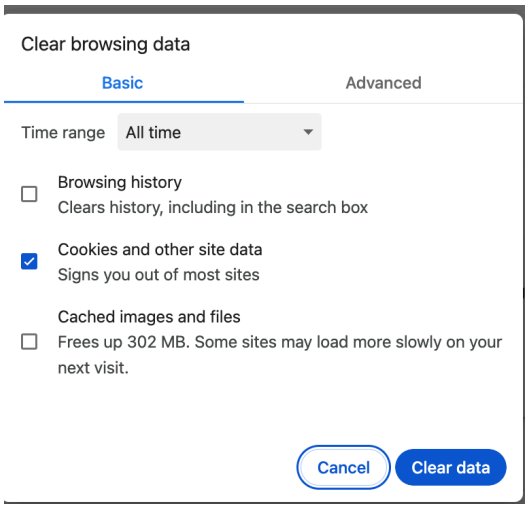
What difficulties have your encountered when installing the extension?

Clear Browsing History

Next, we will ask you to clear your cookie data. Please navigate to chrome://settings/privacy?search=clear (copy and paste the address directly on your search bar), and click on "Clear Browsing Data". Then select **only 'cookies and other site data'**, and click clear data.



Select the time range to be **"All Time"** and select the cookies and other data check box, as seen below. Then click "Clear Data."



Were you able to clear your cookie data?

Yes

No

Intro to website navigation

Now that you have *Cookie Manager* installed, we will ask you to visit a list of 20 websites. Please wait until a banner shows up for each site and interact with the banner as you normally would. We will ask you to answer a few questions after each visit. After you finish the survey task, the frequency of pop-ups will drastically decrease.

Note: for your browsing action to be correctly registered in our database, please directly left-click on the link on the survey page to navigate to the website. If instead you right-click on the link and select "open on a new tab", a warning will continue showing up, meaning that our database has not recognized your click action.

YouTube

Please use Chrome to navigate to youtube.com. Please wait until a banner shows up. Search for a video of your choice.
You haven't clicked on the link

Do you normally visit Youtube?

Yes
No

Have you ever heard of Youtube?

Yes
No

How often do you normally visit Youtube?

At least once a day
At least once a week
Less than once a week
Never

New York Times

Please use Chrome to navigate to nytimes.com. Please wait until a banner shows up. Click on an article of your choice.
You haven't clicked on the link

Do you normally visit NYTimes?

Yes
No

Have you ever heard of New York Times?

Yes
No

How often do you normally visit New York Times?

At least once a day
At least once a week
Less than once a week
Never

Apple Insider

Please use Chrome to navigate to appleinsider.com. Please wait until a banner shows up. Click on an article of your choice.
You haven't clicked on the link

Do you normally visit Apple Insider?

Yes
No

Have you ever heard of Apple Insider?

Yes
No

How often do you normally visit Apple Insider?

At least once a day
At least once a week
Less than once a week
Never

Yahoo

Please use Chrome to navigate to yahoo.com. Please wait until a banner shows up. Click on an article of your choice.
You haven't clicked on the link

Do you normally visit Yahoo?

Yes
No

Have you ever heard of Yahoo?

Yes
No

How often do you normally visit Yahoo?

At least once a day
At least once a week
Less than once a week
Never

Amazon

Please use Chrome to navigate to amazon.com. Please wait until a banner shows up. Search for a product of your choice.
You haven't clicked on the link

Do you normally visit Amazon?

Yes
No

Have you ever heard of Amazon?

Yes
No

How often do you normally visit Amazon?

At least once a day
At least once a week
Less than once a week
Never

eBay

Please use Chrome to navigate to ebay.com. Please wait until a banner shows up. Search for a product of your choice.
You haven't clicked on the link

Do you normally visit eBay?

Yes
No

Have you ever heard of eBay?

Yes
No

How often do you normally visit eBay?

At least once a day
At least once a week
Less than once a week
Never

What is 6 divided by 2? Regardless of the correct answer, you should always select the option with the value "one". This is an attention check question.

2
3
1

Target

Please use Chrome to navigate to target.com. Please wait until a banner shows up. Search for a product of your choice.
You haven't clicked on the link

Do you normally visit Target?

Yes
No

Have you ever heard of Target?

Yes
No

How often do you normally visit Target?

At least once a day
At least once a week
Less than once a week
Never

Etsy

Please use Chrome to navigate to etsy.com. Please wait until a banner shows up. Search for a product of your choice.
You haven't clicked on the link

Do you normally visit Etsy?

Yes
No

Have you ever heard of Etsy?

Yes
No

How often do you normally visit Etsy?

At least once a day
At least once a week
Less than once a week
Never

Turo

Please use Chrome to navigate to turo.com. Please wait until a banner shows up. Click on a car of your choice.
You haven't clicked on the link

Do you normally visit Turo?

Yes
No

Have you ever heard of Turo?

Yes
No

How often do you normally visit Turo?

At least once a day
At least once a week
Less than once a week
Never

StockX

Please use Chrome to navigate to stockx.com. Please wait until a banner shows up. Search for a product of your choice.
You haven't clicked on the link

Do you normally visit StockX?

Yes
No

Have you ever heard of StockX?

Yes
No

How often do you normally visit StockX?

At least once a day
At least once a week
Less than once a week
Never

ESPN

Please use Chrome to navigate to espn.com. Please wait until a banner shows up. Click on an article of your choice.
You haven't clicked on the link

Do you normally visit ESPN?

Yes
No

Have you ever heard of ESPN?

Yes
No

How often do you normally visit ESPN?

At least once a day
At least once a week
Less than once a week
Never

Facebook

Please use Chrome to navigate to facebook.com. Please wait until a banner shows up. Scroll down.
You haven't clicked on the link

Do you normally visit Facebook?

Yes
No

Have you ever heard of Facebook?

Yes
No

How often do you normally visit Facebook?

At least once a day
At least once a week
Less than once a week
Never

Funny Or Die

Please use Chrome to navigate to funnyordie.com. Please wait until a banner shows up. Click on an article of your choice.
You haven't clicked on the link

Do you normally visit Funny Or Die?

Yes
No

Have you ever heard of Funny Or Die?

Yes
No

How often do you normally visit Funny Or Die?

At least once a day
At least once a week
Less than once a week
Never

Weather

Please use Chrome to navigate to weather.com. Please wait until a banner shows up. Search for a location.
You haven't clicked on the link

Do you normally visit Weather.com?

Yes
No

Have you ever heard of Weather.com?

Yes
No

How often do you normally visit Weather.com?

At least once a day
At least once a week
Less than once a week
Never

DuckDuckGo

Please use Chrome to navigate to duckduckgo.com. Please wait until a banner shows up. Search for a product of your choice.
You haven't clicked on the link

Do you normally visit DuckDuckGo?

Yes
No

Have you ever heard of DuckDuckGo?

Yes
No

How often do you normally visit DuckDuckGo?

At least once a day
At least once a week
Less than once a week
Never

Truewerk

Please use Chrome to navigate to truewerk.com. Please wait until a banner shows up and navigate to an item.
You haven't clicked on the link

Do you normally visit Truewerk?

Yes
No

Have you ever heard of Truewerk?

Yes
No

How often do you normally visit Truewerk?

At least once a day
At least once a week
Less than once a week
Never

Thomann

Please use Chrome to navigate to thomannmusic.com. Please wait until a banner shows up and navigate to an item.
You haven't clicked on the link

Do you normally visit Thomann Music?

Yes
No

Have you ever heard of Thomann Music?

Yes
No

How often do you normally visit Thomann Music?

At least once a day
At least once a week
Less than once a week
Never

MerrySky

Please use Chrome to navigate to merrysky.com. Please wait until a banner shows up and search for a location.
You haven't clicked on the link

Do you normally visit Merry Sky?

Yes
No

Have you ever heard of Merry Sky?

Yes
No

How often do you normally visit Merry Sky?

At least once a day
At least once a week
Less than once a week
Never

Seattle Times

Please use Chrome to navigate to seattletimes.com. Please wait until a banner shows up and then click on an article.
You haven't clicked on the link

Do you normally visit Seattle Times?

Yes
No

Have you ever heard of Seattle Times?

Yes
No

How often do you normally visit Seattle Times?

At least once a day
At least once a week
Less than once a week
Never

Semafor

Please use Chrome to navigate to semafor.com. Please wait until a banner shows up and then click on an article.
You haven't clicked on the link

Do you normally visit Semafor?

Yes
No

Have you ever heard of Semafor?

Yes
No

How often do you normally visit Semafor?

At least once a day
At least once a week
Less than once a week
Never

Favorite website

Navigate to your favorite e-commerce website. Please wait until a banner shows up. Search for a product of your choice.

Paste the URL of the product below:

Did you see a cookie consent banner?

Yes
No

Did you make a choice on whether to allow for cookie sharing?

Yes, I allowed my preferred cookies and blocked unwanted cookies
Yes, I chose the default cookie sharing
No, I closed the cookie consent banner
No, I left the website without interacting with the consent banner

Questionnaire

Think about your browsing experiences on a typical day. Overall, how frequently do you encounter cookie consent banners?

Too frequently
A bit more frequently than ideal
Just right
A bit less frequently than ideal
Too infrequently

Overall, how would you rate the ease of navigation of the cookie consent interfaces on the websites you visit?

Very easy to navigate

Moderately easy to navigate

Neither easy nor hard to navigate

Moderately hard to navigate

Very hard to navigate

Overall, how would you rate the ease of making your preferred choices regarding cookie sharing on the websites you visit?

Very easy

Moderately easy

Neither easy nor hard

Moderately hard

Very hard

Which of the following best describes your behavior when deciding whether to share cookies online?

I consider both the website that is asking and the types of cookies involved before deciding whether to share them

I decide whether to share cookies based on what type of cookies they are

I decide whether to share cookies based on which website is asking

I accept most cookies

I reject most cookies

Part1-conclude

Thank you! To finish the rest of the study, we ask you to keep Cookie Manager installed for another seven days. You can continue your browsing activities as usual during this time. The frequency of pop-ups will drastically decrease over time. After the seven days have passed, the extension will prompt you with a survey and the instructions on how to uninstall the extension.

There is no completion code, since our system will detect completion automatically. Please make sure to click the next button below so that we register your response.

Powered by Qualtrics

Intro Page

Thank you for finishing our web browsing task! Now we will walk you through the uninstallation process of the browser extension. To complete the study, we just need to ask you a few more questions about the web browsing and cookie-sharing experiences while using our extension and in general.

Block 1

Think back about your browsing experiences after completing our 20-website visit task while Cookie Manager is installed. Overall, what do you think of the frequency with which cookie consent banners appear during that time?

Too frequent
A bit more frequent than ideal
Just right
A bit less frequent than ideal
Too infrequent

Block 2

Overall, how will you rate the ease of navigation of the cookie consent interface created by our browser extension?

Very easy to navigate
Moderately easy to navigate
Neither easy nor hard to navigate
Moderately hard to navigate
Very hard to navigate

Block 3

Overall, how will you rate the ease of making your preferred cookie sharing choices created by our browser extension?

Very easy
Moderately easy
Neither easy nor hard
Moderately hard
Very hard

Block 4

In the past week, which of the following statement best describes your behavior when deciding whether to share cookies online?

I accepted most cookies

I rejected most cookies

I chose whether to share cookies based on which website is asking

I chose whether to share cookies based on what types of cookie it is

I chose whether to share cookies based on what website is asking and what types of cookie it is

Why choice

Think back to a case when you accepted all cookies during the course of the study. Why did you do so?

Think back to a case when you chose **not** to accept all cookies during the course of the study. Why did you do so?

Block 5

Overall, how do you think the Cookie Manager extension changes your web browsing experience?

It improves my browsing experience by a lot

It improves my browsing experience slightly

It neither improves nor degrades my browsing experience

It degrades my browsing experience slightly

It degrades my browsing experience a lot

Block 12

Consider the cookie consent form below.

One option is to hit the 'x' button in the upper right. If you were to click this 'x', what do you think will happen?

All cookies are accepted.

None of the cookies are accepted.

Other, please explain:

Block 8

During the study period, did you take any actions to change how you browse the internet?

No

Yes, I used a different browser or device.

Yes, I browsed the internet less.

Yes, I did something else. Please specify.

Block 9

As you browse the internet, which information do you think advertisers have about you? Check all that apply.

Your demographic information

Your prior website visits

Your interests

Your prior purchases

Your social media posts

Your address

Your credit score

Block 10

Thinking about privacy polices you might come across online or on your smartphone. Which of the following comes closer to your view, even if neither is exactly right?

Just something I have to get past in order to use a product or service.

A meaningful part of my decision to use a product or service.

Privacy means different things to different people today. In thinking about all of your online browsing, please state how important it is for you to be in control of who can get info about you.

Not all imporant	Not very imporant	Somewhat Important	Very Imporant
1	2	3	4
○	○	○	○

Block 11

What do you think are the benefits of sharing the data listed above with the advertisers?

What do you think are the costs of sharing the data listed above with the advertisers?

Block 6

Do you have any suggestions to help us improve the design of the Cookie Manager extension or the design of our study in general?

Block 14

Consider a tool that that allows you to specify how you would like to answer cookie consent questions online. This tool will then automatically hide all cookie pop-ups and answer them in they way you specified. For example, if you stated that you wanted to accept cookies for all websites, the tool would do so.

Please select how much better or worse the tool is than manually answering the cookie consent form for each website.

Much worse than manual choice	Worse	Somewhat worse	Similar	Somewhat better	Better	Much better than manual choice
○	○	○	○	○	○	○

How much would you be willing to pay for the tool?

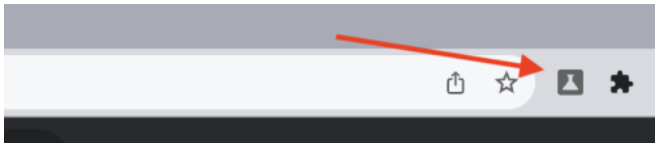
Please enter the price in the text box below.

Instructions for how to download and configure the tool, called Consent-O-Matic, are available here.

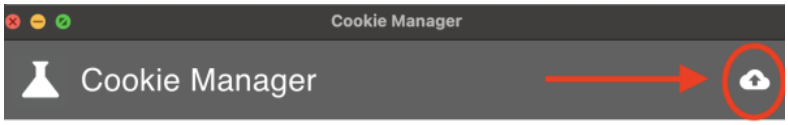
Please click the arrow below to continue the survey.

Block 7

Thank you! We will ask you to upload your data to us prior to uninstalling the extension. Please click on the Cookie Managert extension icon in your Chrome browser.



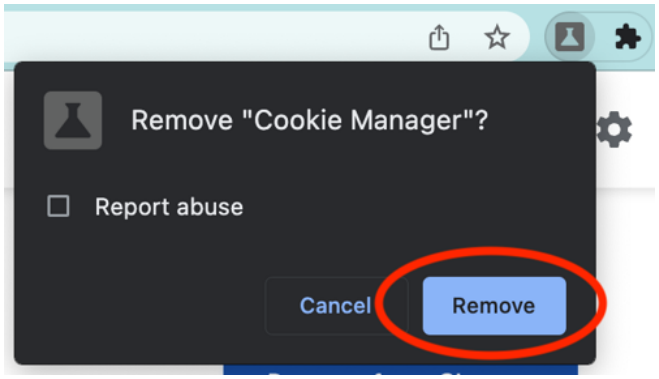
You should see a pop-up. Please click on the cloud button with an arrow. Completing this step ensures that your participation in our study and the associated data are properly recorded.



Now that you've clicked the cloud button, you can now proceed to uninstall the extension. Completing this step ensures that we stop collecting your browsing data going forward.

Here's how you can remove the cookie manager extension:
- Click here.
- Click “Remove from Chrome.”

- Confirm by clicking "Remove" on the pop-up window appearing on the top right corner of your browser.



- You're all set.

If you have trouble uninstalling Cookie Manager, email us at cookie.manager.study@gmail.com and we will help you with additional instructions.

Please click the arrow below to finish the survey.

Powered by Qualtrics

Consent Form

Study: Online Cookie Consent Choices
Lead Researcher: Chiara Farronato

What's This About?

We're a team of researchers from Harvard University and Boston University, exploring how people make privacy decisions online. Our work focuses on understanding how the internet shapes everyday choices and impacts society.

What's Involved?

You'll be asked a few short demographic questions, followed by some questions about your preferences for setting cookie consent online. This will not take more than 6 minutes.

Participants who successfully complete the survey will be compensated $1.20 through Prolific. Please note that some participants may not be eligible to continue based on screening questions or attention checks. In those cases, the survey will end early and compensation will not be provided.

Thinking About Joining the Research Study? Here's What You Should Know!

✨ You're in control: It's 100% your choice whether or not to participate.
💁‍♀️ Totally voluntary: You can say "yes" or "no" with no pressure at all.
🔄 Change your mind? No problem: You can join and later decide to stop, anytime.
🛑 No penalties: Saying "no" won't affect your rights, benefits, or how you're treated.
❓ Got questions? Ask away! You can (and should!) ask all the questions you need to feel comfortable before deciding.

Your Choice, Your Terms

Participation is 100% voluntary. You're free to opt in or out at any time, and your decision won't affect you in any way. Feel free to ask questions before you begin. If you proceed, we ask that you complete all questions.

Who can I talk to?

If you have any questions, concerns, or complaints, or think the research has hurt you, talk to the research team at chaskin@hbs.edu.

ADDENDUM FOR DATA SUBJECT TO NON-U.S. DATA PRIVACY LAWS:
This Addendum concerns requirements for data which is subject to various non-U.S. privacy laws. If you are participating in this study in a country other than the United States ("your country"), your country may have a law with certain requirements for processing personal data relating to identified or identifiable individuals located in your country that applies to entities outside of your country, including entities in the U.S. This Addendum will refer to each of those laws as the "Law."
Harvard researchers, led by Prof. Chiara Farronato (the "study team") will collect information about you during the course of the study. This form calls such information your "Personal Information" and it will include your responses to the survey below. No information that can directly or indirectly identify you will be collected.
Harvard is the controller of your Personal Information collected for this study. By agreeing to participate below, you consent to the study team's use and processing of your Personal Information for the purpose of this study. The study team, and those working with the study team, will process your Personal Information in compliance with applicable Law.
If you withdraw your permission, you will no longer be able to participate in the study, and no new information will be collected about you or from you by the study team. However, if you withdraw, the study team will retain the Personal Information collected prior to your withdrawal and may use it to guarantee the integrity of the study and for any other purpose permitted under the Law. Withdrawal will not affect the lawfulness of the data processing that occurred prior to your withdrawal.
By agreeing below, you consent to the transfer of your Personal Information to countries other than your country, including the United States. The study team may process your Personal Information in countries other than your country. These countries may have established standards and practices for managing Personal Information that differ from those of your country.

Regulators in your country may have determined that the data protection laws of the US and certain other countries do not protect Personal Information to the same extent as your country. Harvard, and those identified in this Addendum as working with Harvard, will take steps to maintain the confidentiality of your Personal Information.

Harvard may require access to your Personal Information to assist in conducting and managing this study, and to comply with legal or regulatory requirements, including to: verify that the study is conducted correctly and that study data are accurate; answer questions from Institutional Review Board(s) (IRB(s)), Independent Ethics Committee(s) (IEC(s)), or government or regulatory agencies.

The study team may also use your Personal Information to:

• Publish summaries of the study results in academic journals, on the internet, or at educational meetings of other researchers. You will not be directly identified in any publication or report of the study. However, some journal representatives may need access to your Coded Information to verify the study results and ensure the research meets the journal's quality standards. Also, journals may require that certain data from the study that does not directly identify you (i.e., de-identified survey responses) be made available to other researchers for further research projects.

• Improve the quality, design, and safety of this study and other research studies.

• Conduct additional studies with the data collected in this study to advance scientific research. At this time, we do not know the specific details of these future research projects.

Under your country's Law, you may have certain rights related to your Personal Information collected under this study. If you wish to exercise those rights, please contact EEAdatasubjectrequest@harvard.edu. Please note that under your country's Law, you may also have the right to make a complaint to your country's data protection authority, if you so choose.

Under GDPR:

• You have the right to see the information being collected about you in the study.

• You have the right to correct or update your Personal Information if it is inaccurate.

• You have the right to limit the collection and use of your Personal Information under certain circumstances (for example, if you think that the information is inaccurate).
• You have the right to receive your Personal Information in a structured, common computer format (for example, in a readable text electronic file or chart) for your own purposes or for giving it to others, as required by applicable data protection laws. You may not have the right to receive your Personal Information that has been used for public interest purposes or in the exercise of official authority vested in Harvard.
• You have the right to request the deletion of your Personal Information if you are no longer participating in the study. However, there are limits on your ability to request deletion of your Personal Information. Harvard may keep and use some or all of your Personal Information if deletion would seriously impair the study (for example, if deletion would affect the consistency of study results) or if your Personal Information is needed to comply with legal requirements.

Ready to share your insights?
Please indicate below if you agree to participate.

- ○ I agree to participate
- ○ I do not agree to participate

No Consent

Thank you for letting us know that you do not want to participate.

Demographics

What is your Prolific ID?

Where do you currently live?

What is your age?

- Less than 18 years old
- 18-29 years old
- 30-39 years old
- 40-49 years old
- 50-59 years old
- 60 years old or older

Which of the following best describes your race or ethnicity?

- White
- Hispanic or Latinx
- Black or African American
- Asian or Pacific Islander
- Other. Please explain:

How many people live in your household?

- Only myself
- 2 people
- 3 people
- 4 people
- 5 or more people

How many children under the age of 18 live in your household?

- None
- 1 child
- 2 children
- 3 or more children

Maria went to the store to buy 4 apples. How many apples did Maria buy? Regardless of the answer, please select the number three.

- 1
- 2
- 3
- 4
- 5 or more

What is the highest level of school you have completed?

- Some high school
- High school degree
- Associate's degree in college
- Bachelor's degree in college (BA, AB, BS,...)
- Graduate degree (MA, MS, MBA, MD, JD, PhD,...)

What was your total household income (in €) in 2024? Please include only employment income (wages, salary, bonuses, tips, and any income from your own businesses).

- €0 to €9,999
- €10,000 to €19,999
- €20,000 to €29,999
- €30,000 to €49,999
- €50,000 to €74,999
- €75,000 to €99,999
- €100,000 and up

What was your total household income (in $) in 2024? Please include only employment income (wages, salary, bonuses, tips, and any income from your own businesses).

- $0 to $24,999

- ○ $25,000 to $49,999
- ○ $50,000 to $74,999
- ○ $75,000 to $99,999
- ○ $100,000 to $149,999
- ○ $150,000 to $199,999
- ○ $200,000 and up

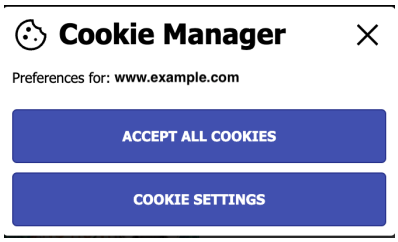
Not Eligible - demographics

Thank you for your response! Unfortunately, you do not qualify to participate in our survey. This may be because you do not fall within the demographic group we are targeting or because an attention check was not passed.

Dark Patterns 1

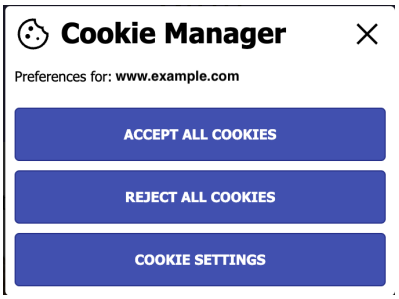
Websites often ask visitors to make choices about cookies (small pieces of data stored in your browser). Below, you will see two different cookie consent banners. Please imagine you visit a website and encounter a banner that asks you to make cookie consent choices.

Which of these two banners would you prefer?



Banner A

○



Banner B

○

How much would you be willing to pay per month for all the consent banners you encounter online to look like **Banner A** instead of **Banner B**?

Enter the amount in your local currency. Use only numbers (decimals allowed), without symbols or words.

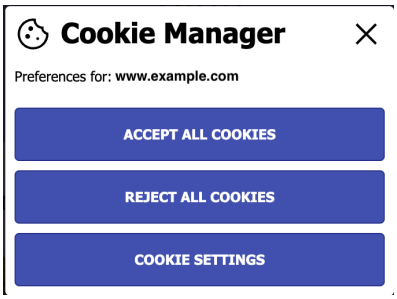
How much would you be willing to pay per month for all the consent banners you encounter online to look like **Banner B** instead of **Banner A**?

Enter the amount in your local currency. Use only numbers (decimals allowed), without symbols or words.

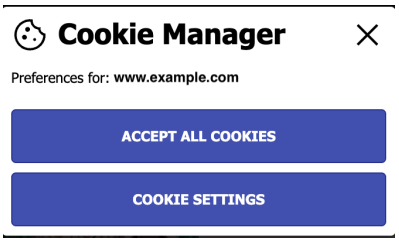
Dark Patterns 2

Websites often ask visitors to make choices about cookies (small pieces of data stored in your browser). Below, you will see two different cookie consent banners. Please imagine you visit a website and encounter a banner that asks you to make cookie consent choices.

Which of these two banners would you prefer?



Banner A



Banner B

How much would you be willing to pay per month for all the consent banners you encounter online to look like **Banner A** instead of **Banner B**?

Enter the amount in your local currency. Use only numbers (decimals allowed), without symbols or words.

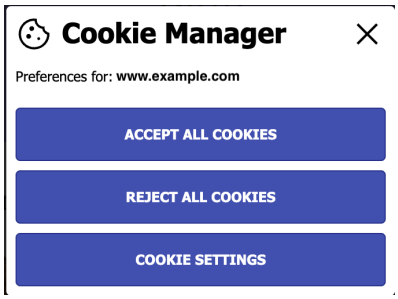
How much would you be willing to pay per month for all the consent banners you encounter online to look like **Banner B** instead of **Banner A**?

Enter the amount in your local currency. Use only numbers (decimals allowed), without symbols or words.

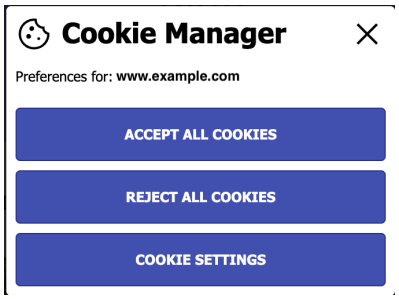
Attention Check

Below are two additional cookie consent banners. As before, imagine you visit a website and encounter a banner that asks you to make cookie consent choices.

Which of these two banners would you prefer? Regardless of your answer, please select Banner B.



Banner A



Banner B

Not Eligible

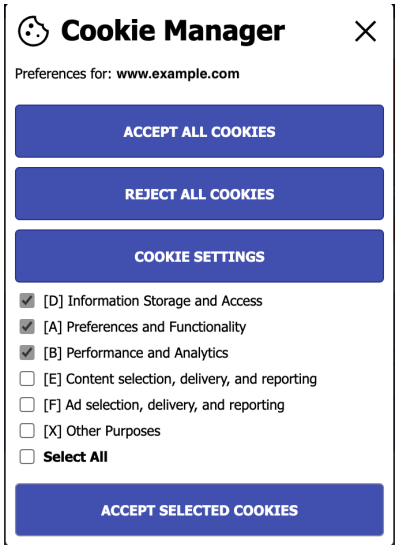
Thank you for your response! Unfortunately, you do not qualify to participate in our survey. This is because an attention check was

not passed.

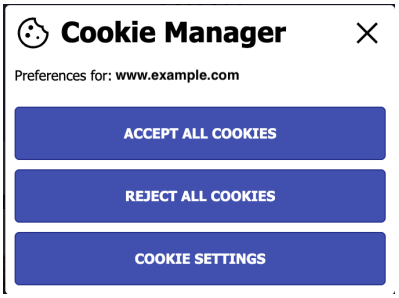
Granular Choices 1

Below are two additional cookie consent banners. As before, imagine you visit a website and encounter a banner that asks you to make cookie consent choices.

Which of these two banners would you prefer?



Banner A



Banner B

How much would you be willing to pay per month for all the consent banners you encounter online to look like **Banner A** instead of **Banner B**?

Enter the amount in your local currency. Use only numbers (decimals allowed), without symbols or words.

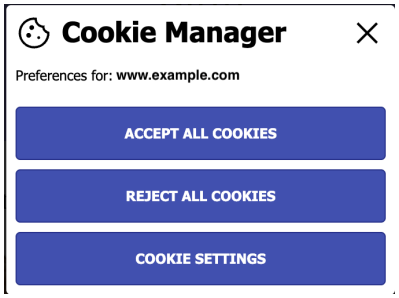
How much would you be willing to pay per month for all the consent banners you encounter online to look like **Banner B** instead of **Banner A**?

Enter the amount in your local currency. Use only numbers (decimals allowed), without symbols or words.

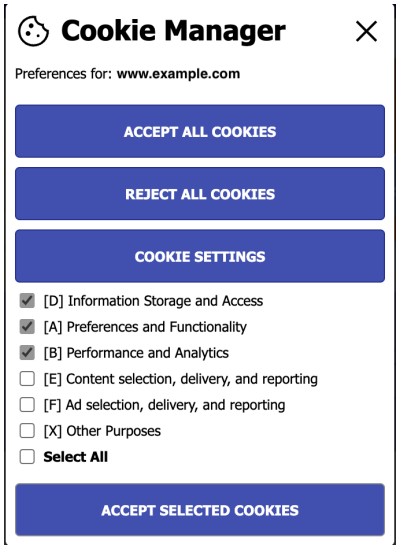
Granular Choices 2

Below are two additional cookie consent banners. As before, imagine you visit a website and encounter a banner that asks you to make cookie consent choices.

Which of these two banners would you prefer?



Banner A



Banner B

How much would you be willing to pay per month for all the consent banners you encounter online to look like **Banner A** instead of **Banner B**?

Enter the amount in your local currency. Use only numbers (decimals allowed), without symbols or words.

How much would you be willing to pay per month for all the consent banners you encounter online to look like **Banner B** instead of **Banner A**?

Enter the amount in your local currency. Use only numbers (decimals allowed), without symbols or words.

Browser Level Consent

Consider a tool that allows you to specify how you would like to answer cookie consent questions online, then apply your preference to all websites while automatically hiding all cookie consent banners. For example, if you stated that you wanted to accept cookies for all websites, the tool would automatically accept all cookies for every website you visit.

Please select how much better or worse this tool is than manually answering your ideal cookie consent banner for each website.

Much worse than manual choice	Worse	Somewhat worse	Similar	Somewhat better	Better	Much better than manual choice
○	○	○	○	○	○	○

According to the description above, what does the tool do automatically?

- ○ It stores your passwords
- ○ It hides cookie banners and answers them based on your preferences
- ○ It blocks all websites completely
- ○ It buys products online for you

How much would you be willing to pay per month for this tool instead of having consent banners for each website you visit?

Enter the amount in your local currency. Use only numbers (decimals allowed), without symbols or words.

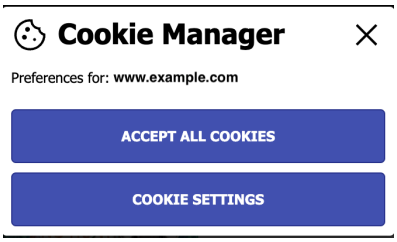
How much would you be willing to pay per month for consent banners for each website you visit instead of having this tool?

Enter the amount in your local currency. Use only numbers (decimals allowed), without symbols or words.

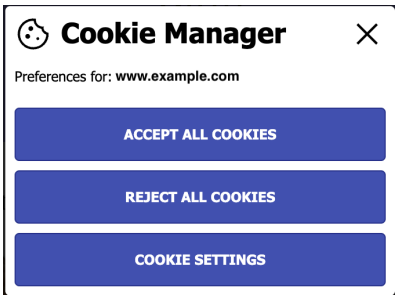
Dark Patterns WTA 1

Websites often ask visitors to make choices about cookies (small pieces of data stored in your browser). Below, you will see two different cookie consent banners. Please imagine you visit a website and encounter a banner that asks you to make cookie consent choices.

Which of these two banners would you prefer?



Banner A



Banner B

How much compensation would you require per month to accept that all the consent banners you encounter online look like **Banner B** instead of **Banner A**?

Enter the amount in your local currency. Use only numbers (decimals allowed), without symbols or words.

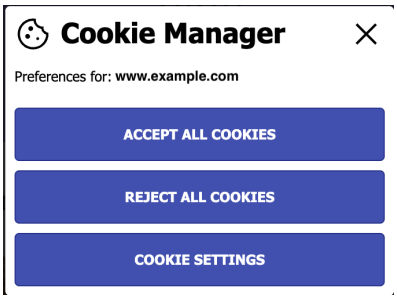
How much compensation would you require per month to accept that all the consent banners you encounter online look like **Banner A** instead of **Banner B**?

Enter the amount in your local currency. Use only numbers (decimals allowed), without symbols or words.

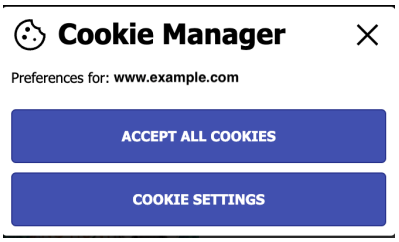
Dark Patterns WTA 2

Websites often ask visitors to make choices about cookies (small pieces of data stored in your browser). Below, you will see two different cookie consent banners. Please imagine you visit a website and encounter a banner that asks you to make cookie consent choices.

Which of these two banners would you prefer?



Banner A



Banner B

How much compensation would you require per month to accept that all the consent banners you encounter online look like **Banner B** instead of **Banner A**?

Enter the amount in your local currency. Use only numbers (decimals allowed), without symbols or words.

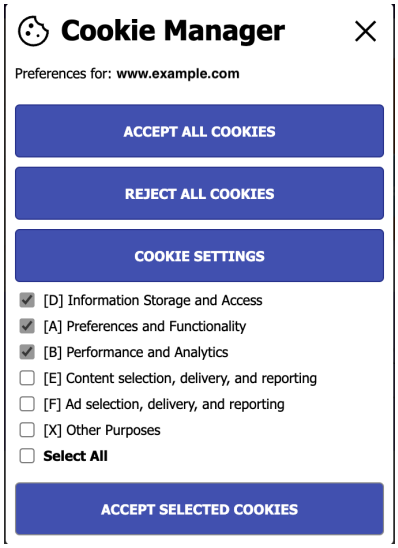
How much compensation would you require per month to accept that all the consent banners you encounter online look like **Banner A** instead of **Banner B**?

Enter the amount in your local currency. Use only numbers (decimals allowed), without symbols or words.

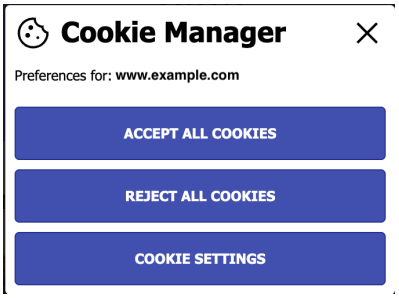
Granular Choices WTA 1

Below are two additional cookie consent banners. As before, imagine you visit a website and encounter a banner that asks you to make cookie consent choices.

Which of these two banners would you prefer?



Banner A



Banner B

How much compensation would you require per month to accept that all the consent banners you encounter online look like **Banner B** instead of **Banner A**?

Enter the amount in your local currency. Use only numbers (decimals allowed), without symbols or words.

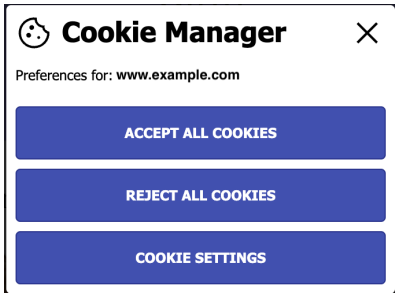
How much compensation would you require per month to accept that all the consent banners you encounter online look like **Banner A** instead of **Banner B**?

Enter the amount in your local currency. Use only numbers (decimals allowed), without symbols or words.

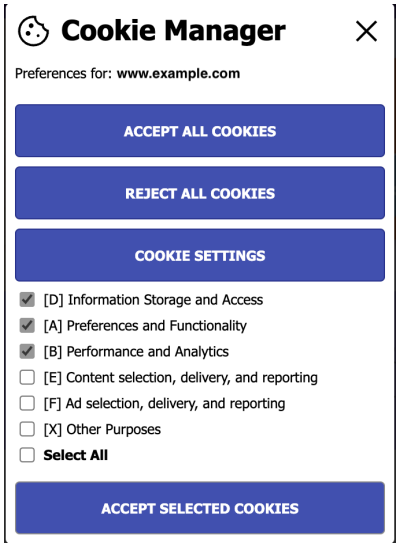
Granular Choices WTA 2

Below are two additional cookie consent banners. As before, imagine you visit a website and encounter a banner that asks you to make cookie consent choices.

Which of these two banners would you prefer?



Banner A



Banner B

How much compensation would you require per month to accept that all the consent banners you encounter online look like **Banner B** instead of **Banner A**?

Enter the amount in your local currency. Use only numbers (decimals allowed), without symbols or words.

How much compensation would you require per month to accept that all the consent banners you encounter online look like **Banner A** instead of **Banner B**?

Enter the amount in your local currency. Use only numbers (decimals allowed), without symbols or words.

Browser Level Consent WTA

Consider a tool that allows you to specify how you would like to answer cookie consent questions online, then apply your preference to all websites while automatically hiding all cookie consent banners. For example, if you stated that you wanted to accept cookies for all websites, the tool would automatically accept all cookies for every website you visit.

Please select how much better or worse this tool is than manually answering your ideal cookie consent banner for each website.

Much worse than manual choice	Worse	Somewhat worse	Similar	Somewhat better	Better	Much better than manual choice
○	○	○	○	○	○	○

According to the description above, what does the tool do automatically?

- ○ It hides cookie banners and answers them based on your preferences
- ○ It blocks all websites completely
- ○ It buys products online for you
- ○ It stores your passwords

How much compensation would you require per month to have consent banners for each website you visit instead of using this tool?

Enter the amount in your local currency. Use only numbers (decimals allowed), without symbols or words.

How much compensation would you require per month to have this tool instead of consent banners for each website you visit?

Enter the amount in your local currency. Use only numbers (decimals allowed), without symbols or words.

Privacy Preferences

Privacy means different things to different people. In thinking about all of your online browsing, please state how important it is

for you to be in control of who can get information about you.

Not at all important	Not very important	Neutral	Somewhat important	Very important
○	○	○	○	○

When a website shows a cookie consent banner, what do you usually do?

- ○ Customize based on the cookie functionality (e.g., analytics, advertising)
- ○ Reject all cookies
- ○ Customize based on both the website and the cookie functionality
- ○ Customize based on the website (trust / brand)
- ○ Close the banner without making a choice
- ○ Accept all cookies

As you browse the internet, which of the following do you think advertisers typically know about you? (Select all that apply)

- ☐ My credit score
- ☐ My contact information (email, phone number, etc.)
- ☐ My national identification number (for example, social security number)
- ☐ Websites I visit and things I search for (browsing history)
- ☐ The device I'm using (phone, computer, etc.)
- ☐ My age, gender, or other demographic information
- ☐ How much time I spend on websites or apps
- ☐ My health information
- ☐ My likes, clicks, and interactions on social media
- ☐ My location
- ☐ [ ] Other. Please explain:
- ☐ I don't know

Successful completion

Thank you for successfully completing the survey! Please click on the arrow below to be redirected back to Prolific to register your submission.

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